

PERFORMANCE OF DISTANCE RELAY WITH DIFFERENT IMPEDANCE CHARACTERISTICS IN THE PRESENCE OF STATCOM FOR VARIOUS FAULTS ON TRANSMISSION LINES

¹Waqas Ahmad Sheikh, ²Farhan Ali, ³Muhammad Ilyas, ⁴Awais Maqsood,
^{*5}Abdul Basit Butt

¹EE Deptt. SEN, University of Management and Technology, Lahore 54770, Pakistan

²School of Computer and IT, Beaconhouse National University, P.O. Box 53700 Lahore, Pakistan

³Department of Software Engineering, Superior University, Lahore 5400 Pakistan

⁴EE Deptt. SEN, University of Management and Technology, Lahore 54770, Pakistan

^{*5}EE Deptt. SEN, University of Management and Technology, Lahore 54770, Pakistan

¹waqasahmad@umt.edu.pk, ^{*2}farhan32748@gmail.com, ³muhammad.ilyas@superior.edu.pk,

⁴awais.maqsood@umt.edu.pk, ^{*5}abdul.basit.butt05@gmail.com

DOI:- <https://doi.org/10.5281/zenodo.20610768>

Article History

Received: 10 May, 2026

Accepted: 02 June, 2026

Published: 03 June, 2026

Copyright @Author

Corresponding Author: *

Abdul Basit Butt

Abstract

The rapid growth of electrical power demand necessitates the construction of transmission lines and power grids without compromising power quality. To achieve high power quality, advancements in semiconductor devices are utilized in the form of Flexible AC Transmission Systems (FACTS) [1]. FACTS devices compensate the system by altering parameters, especially power system impedance and power angle, making it essential to analyze system performance and transmission line protection, as relay settings depend on line impedance. The distance relay is a critical protective relay in this regard. FACTS devices include SVC, TSC, TCR, TCSC, and self-commutated VAR compensators such as STATCOM [2]. SVCs are used for damping power swings and improving stability [3]-[4]. Among FACTS devices, STATCOM is the most versatile, providing both leading and lagging reactive power, fast response, low harmonic content, no inrush current, and good dynamic performance [5]. DSTATCOM is commonly used for reactive power compensation. This work analyzes distance relay performance with different impedance characteristics in the presence of DSTATCOM using MATLAB/Simulink, and a communication-aided scheme ensures correct zone-1 operation without mal-operation.

1. Introduction

The electrical transmission system must be capable of handling the bulk of power provided to end consumers from the generation station [3]. The power system must provide reliable energy to consumers in general and industries in particular. Electrical demand is rapidly growing, and new generating stations and transmission lines are constructed to fulfill these electricity demands. To enhance reliability while transmitting bulk power to end users, different advanced supporting techniques have been introduced. With the ever-increasing demand for electrical power, the transmission system must operate within its thermal limits while supplying energy to the end user.

In [4], DSTATCOM is proposed to alleviate variations of both positive and negative sequence voltages at the fundamental frequency. A current-controlled voltage source inverter with a DC bus capacitor is used as DSTATCOM, and voltage or current control modes are employed [5].

As these FACTS devices compensate the system to achieve required power quality by changing system parameters, especially power system impedance and power angle, it is essential to analyze system element performance based on impedance [6]. Transmission lines in Pakistan pass through diverse areas and harsh environmental conditions, making fault occurrence highly probable. These lines are hundreds of kilometers and interconnected network requires fault analysis for stable bulk power flow. Reliable electrical power protection systems require protective relays with high dependability, accuracy, and speed.

1.1 Statement of Problem

The induction of FACTS devices impacts protection relay performance, making transmission protection more complex. At fault, FACTS devices affect apparent impedance, causing potential relay mal-operation [7]. STATCOM affects distance relay impedance

measurement, phase selection, and operating times [8]. Relay algorithms can be simulated in MATLAB/SIMULINK using complete line parameters. STATCOM models allow zone-wise relay performance analysis. In Pakistan, 500KV, 220KV, and 132KV lines are used, with 500KV and 220KV maintained by NTDCL [9]. These long, interconnected lines face harsh conditions, making fault analysis essential. A 100MVAR STATCOM is proposed on the 220KV Multan-KAPCO-3 line with a D-60 distance relay.

1.2 Literature Survey

Power system stability is crucial for reliable transmission, and FACTS devices enhance controllability and maintain stable power flow. However, these devices affect protection relays, which are sensitive and vital subsystems. FACTS devices' fast response can disturb relay performance. Shunt compensators impact distance relays by causing zone reach selectivity issues, though operating time was not considered [10]. Naresh Patnana et al. [11] studied TCSC and STATCOM on a 400KV line, observing apparent impedance variation in distance relays with Quadrilateral characteristics, without analyzing operating time. UPFC impact on distance relays during power swings showed system impedance and power swing settings were affected [12]. Mohammed Elsamaly et al. [13] studied STATCOM mid-line on a hydro system, showing adverse effects on LOE relays and parallel generators. Danna Hemasunder et al. [14] evaluated STATCOM impact on relay reach; Gorakshanath Abande et al. [15] predicted tripping without analyzing different faults. Mohammad Zellague [16][17] studied GCSC, TCSC, and TCSR devices on 400KV lines, analyzing relay behavior and phase-to-ground faults. Gaber El-Saady et al. [18] studied SSSC and STATCOM; Sankara Subramanian et al. [19] highlighted load angle and resonance effects; Wen-Hao Zhang et al. [20] analyzed 500KV lines, and Mohammad Javad Farah et al. [21] observed

mal-operation of impedance relays with Shunt and Series FACTS devices.

2. FACTS Devices

This chapter discusses FACTS devices and their role in analyzing distance relay performance. Reactive power compensation is explained, and different FACTS devices are compared. To meet growing power demand, system expansion faces economic and environmental limits, making reactive power compensation an effective solution to improve power system stability. The

simple electrical system is shown in figure 2.1, the two-power system having voltage V_1 and V_2 are connected to the transmission line having line Reactance X_L

$V_1 = V_1 \angle \alpha_1$ at Bus – 1 and $V_2 = V_2 \angle \alpha_2$, therefore, transmission angel is given as;

$$\alpha = \alpha_1 - \alpha_2 \quad (2.1)$$

The voltage drops in transmission line is given as;

$$V_{LD} = V_1 - V_2 \quad (2.2)$$

Similarly, current carried out by transmission line is given as;

$$I_L = V_{LD}/X_L \quad (2.3)$$

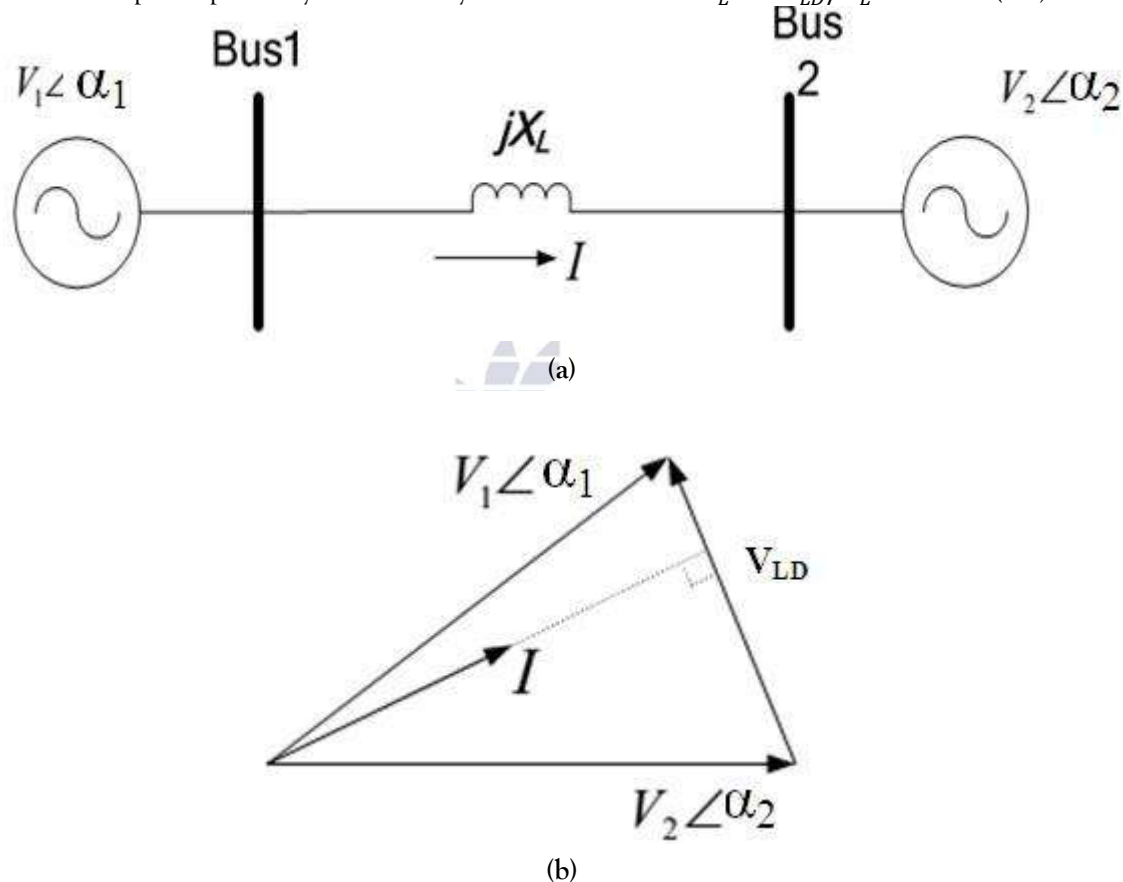


Fig 2.1; Sample Transmission system: (a) Single Line network; (b) Phasor Representation

Current Contributing Reactive Power from sending end;

$$I_{q1} = \frac{V_2 \sin \alpha}{X_L}$$

Current Contributing Active power;

$$I_{p1} = \frac{(V_1 V_2 \cos \alpha)}{X_L}$$

Reactive Power from sending End is;

$$Q_1 = \frac{V_1 - V_2 \sin \alpha}{X_L} \quad (2.4)$$

Active Power flow from sending end;

$$P_1 = \frac{V_1(V_1 - V_2 \cos \alpha)}{X_L} \quad (2.5)$$

Similarly, currents at receiving end are;

$$I_{p2} = \frac{V_2 \sin \delta}{X_L}$$

$$I_{Q2} = \frac{(V_2 - V_1 \cos \delta)}{X_L}$$

The Receiving End Active Power end Bus2 is;

$$P_2 = \frac{V_1 V_2 \sin \delta}{X_L} \quad (2.6)$$

And the Reactive Power;

$$Q_2 = \frac{V_2(V_2 - V_1 \cos \delta)}{X_L} \quad (2.7)$$

It is clear from the above discussion that reactive power is dependent on

- Voltages at both end

- Transmission Angle
- Line reactance

Reactive power transmission is controlled by changing system parameters, each with limitations. Increasing voltage requires more insulation, changing transmission angle raises cost and affects stability. Power flow is efficiently improved by controlling line reactance using series or parallel capacitors [22]. The uncompensated power system with generation feeding a load through a transmission line is shown in Figure 2.2.

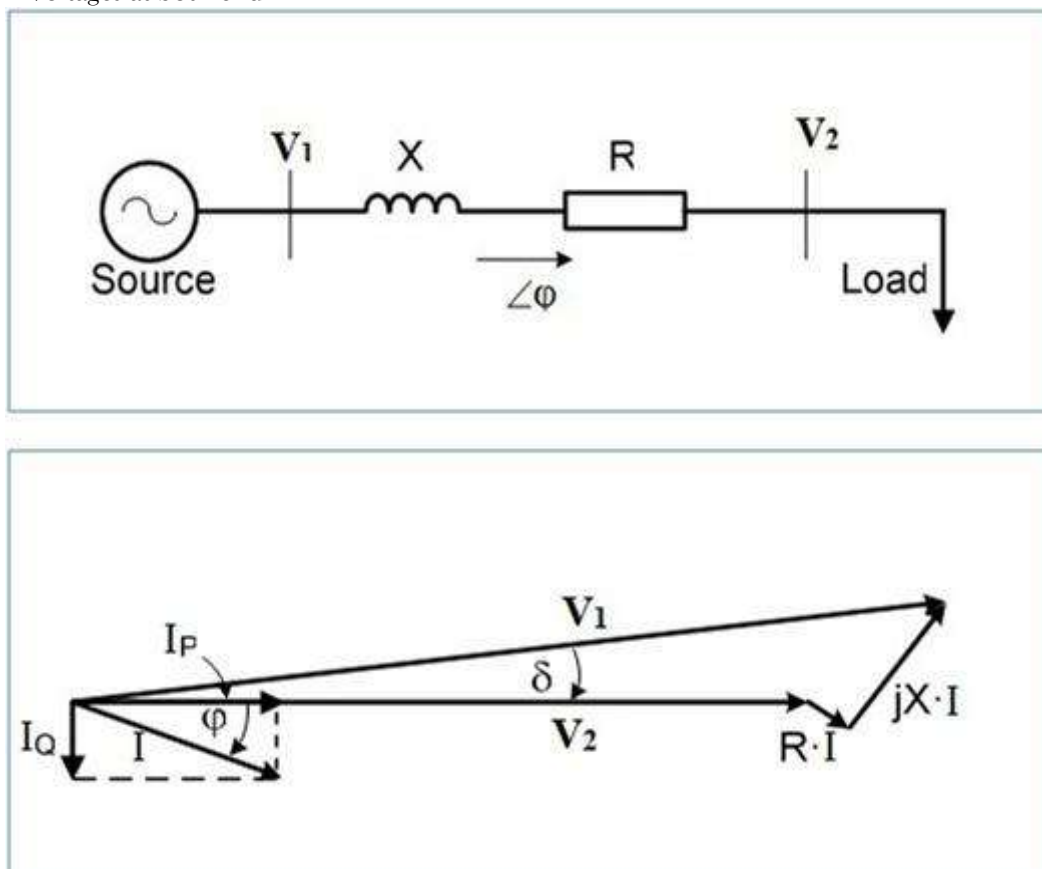


Fig. 2.2: Power System with phasor diagram without compensation.

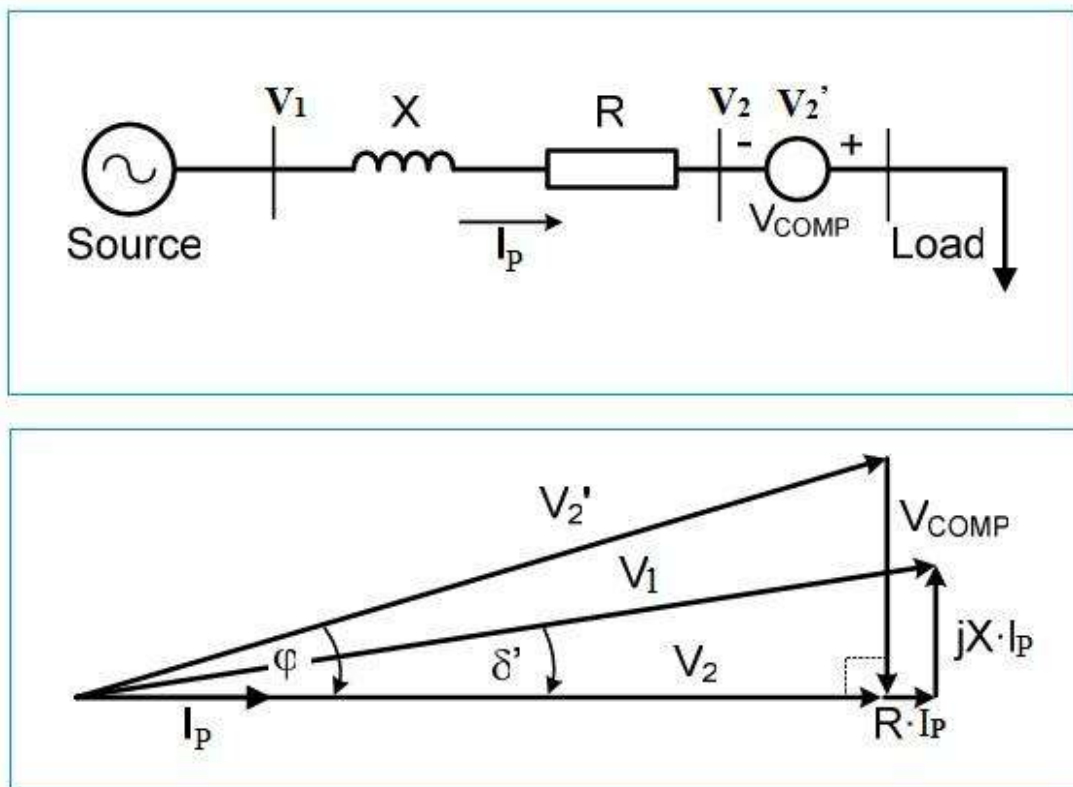


Fig. 2.3: Series Compensated system with a Voltage Source & its Phasor Diagram

Series compensation is introduced in the power system (Figure 2.3), improving power factor toward unity. A system without compensation is shown in Figure 2.4, with source voltage V_1 and load voltage V_2 represented in the phasor diagram.

Institute for Excellence in Education & Research

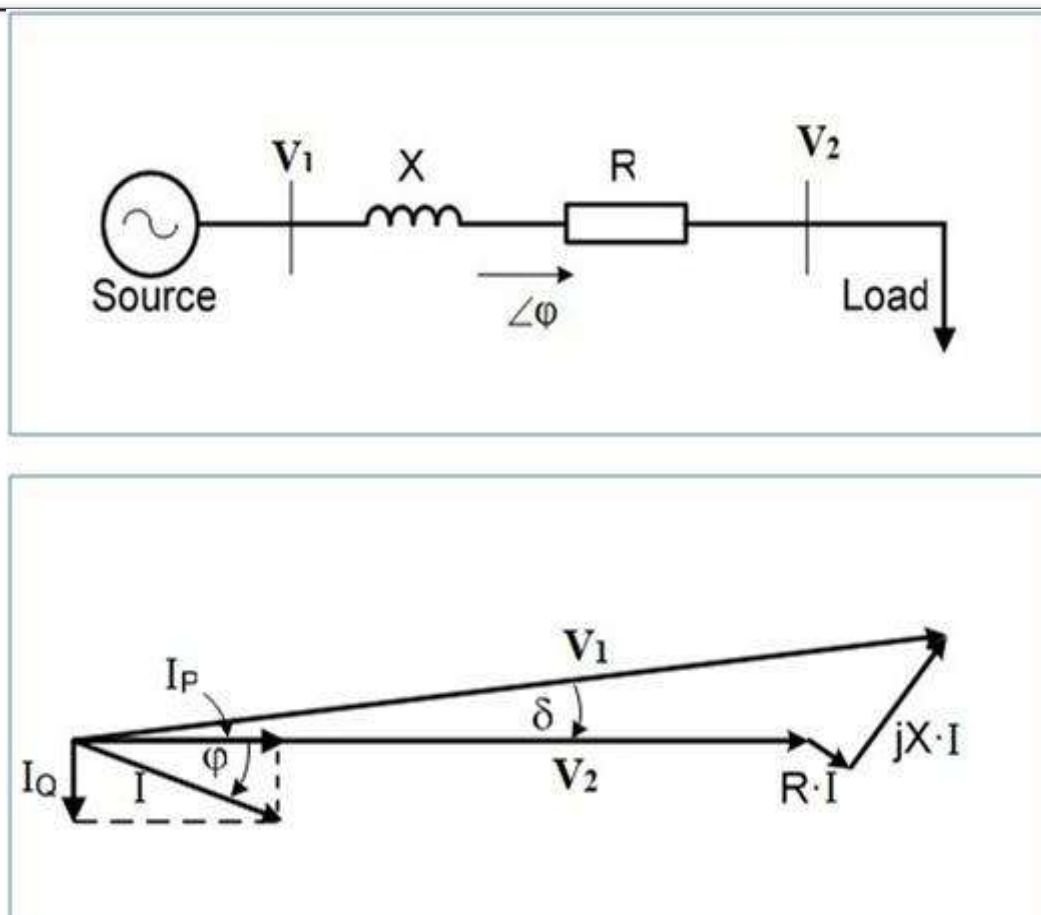


Fig. 2.4: Phasor diagram and Single line of power System without Compensation.

Figure 2.4 shows reactive current depends on load type; inductive loads require reactive current. Supplying it near the load minimizes line losses.

Shunt compensation with a parallel source (Figure 2.5) improves power transfer and reduces source reactive current.

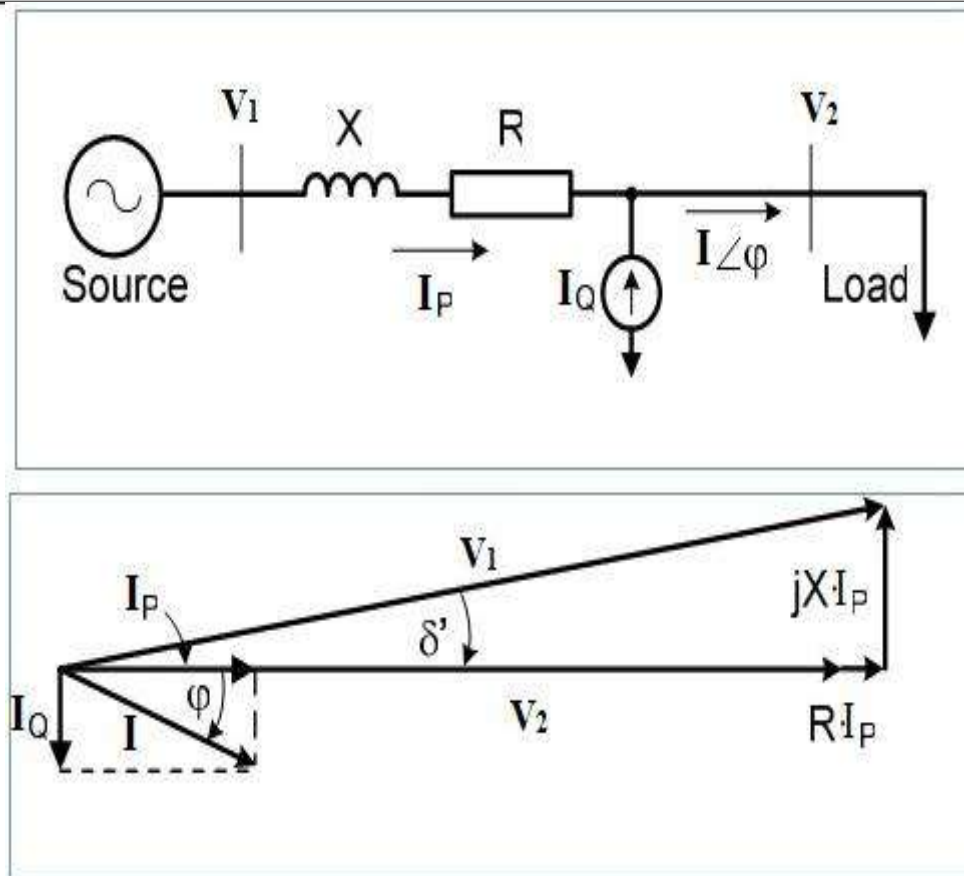


Fig. 2.5: Power system with Shunt Compensation

If the power factor is already leading, compensation can be achieved by connecting a shunt inductor. Advancements in power electronics led to Flexible AC Transmission Systems (FACTS) to improve and control power transfer. FACTS change system parameters—phase angle, impedance, voltage—and can be connected in series or parallel to enhance controllability [23]. They are categorized into two

generations: the first includes generation-end devices like tap-changing transformers, capacitors, and synchronous condensers [24]; the second generation has faster response and is further divided into two groups.

- Devices Based on Voltage Source.
- Thyristor-based devices.

The classification of FACTS devices is shown in figure 2.6,

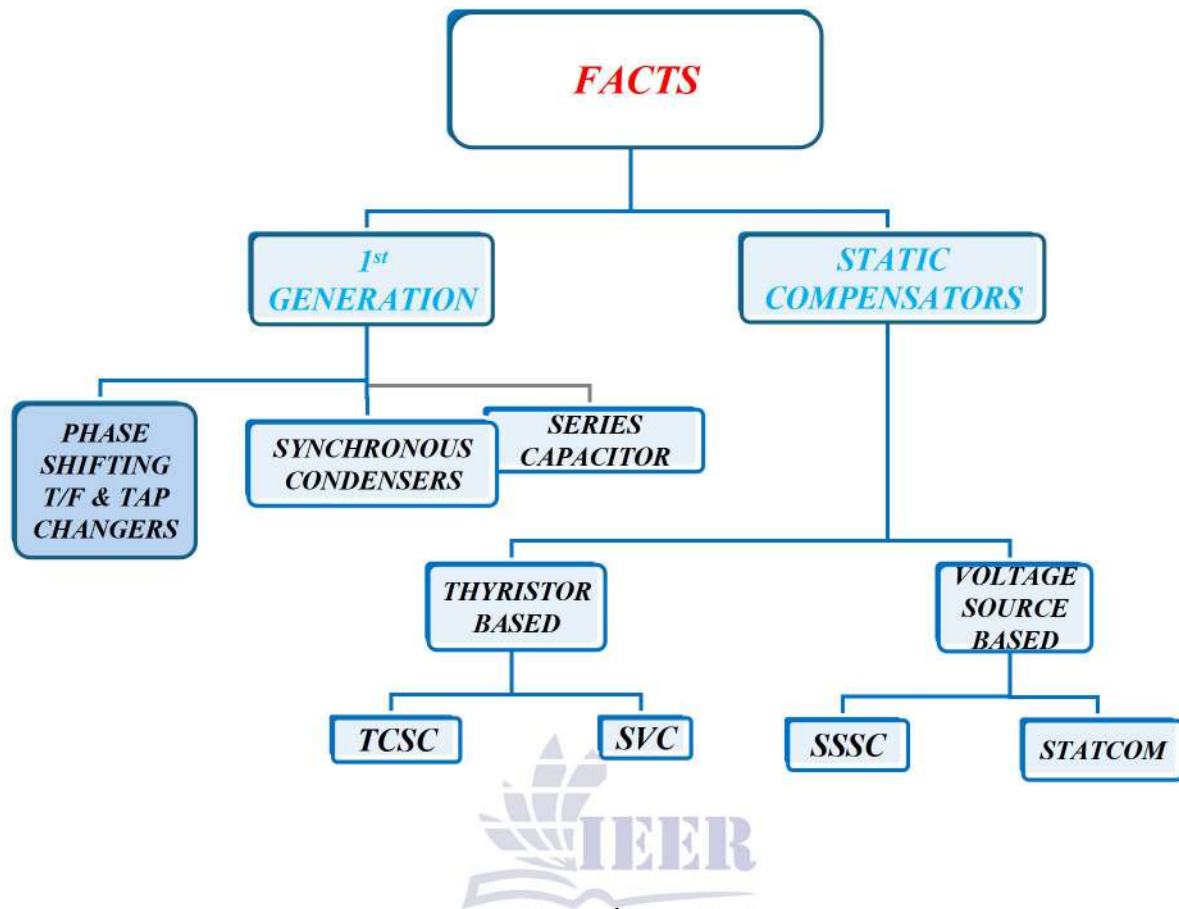


Fig. 2.6: Types of FACTS Devices.

The main thyristors are used with a combination of the inductive and capacitive element to achieve the required function; these are controlled devices and can be classified in series as well as shunt compensation.

For the VAR compensation the SVC is connected in parallel with the system, the SVC exchanges the reactive power it can absorb or inject the reactive power in a system. The SVC is voltage based device and maintains the system

voltage within the desired limits [16]. SVC can be further divided as following;

- TSC-TCR type SVC
- TCR-FC type SVC
- TSC-TSR type SVC

This above classification is based on the configuration of the SVC and discussed as follow. Figure 2.7, represents the TSC-TCR configuration.

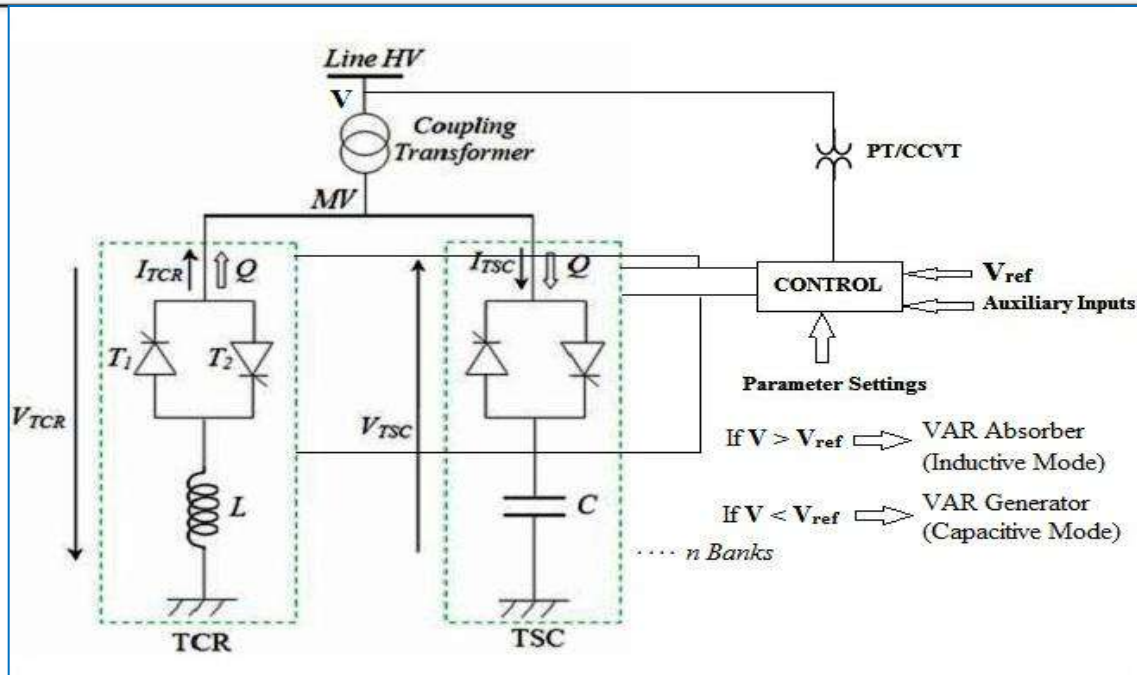


Fig. 2.7: Block Diagram of TSC-TCR type SVC Configuration

TCR provides reactive power by controlling the delay angle [25]. TSC injects stepwise reactive power using multiple capacitors, operating in inductive and capacitive modes. Coordinating TSC with TCR achieves continuous controlled

reactive power. SVC maintains the desired system voltage at the transmission line connection point. The current-voltage relation of SVC is shown in Figure 2.8.

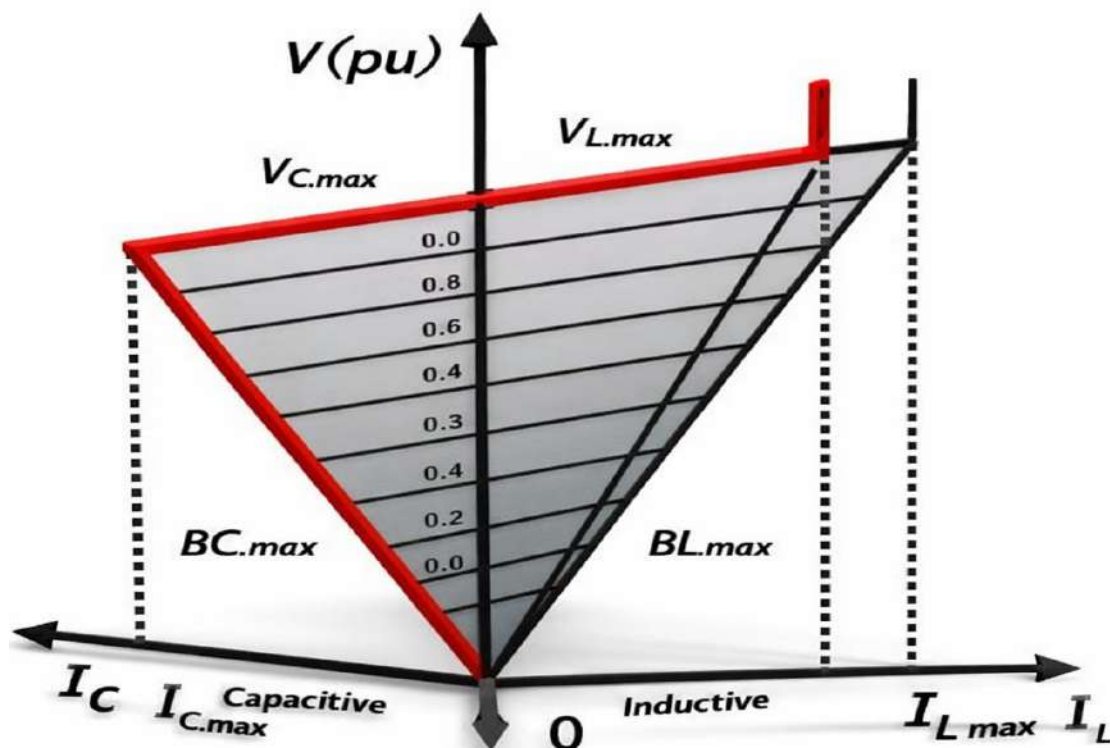


Fig. 2.8: TSC-TCR type SVC Voltage and current Characteristics.

The slope value sets the required system voltage and reactive power [26]. TSC-TRC SVC can act as a fixed inductor or capacitor at maximum output, reducing voltage control. In TCR-FC SVC, the thyristor firing angle regulates voltage, with a parallel fixed capacitor controlling lagging VARs and FC MVA lower than TCR MVA. The operating principle is shown in Figure 2.9.

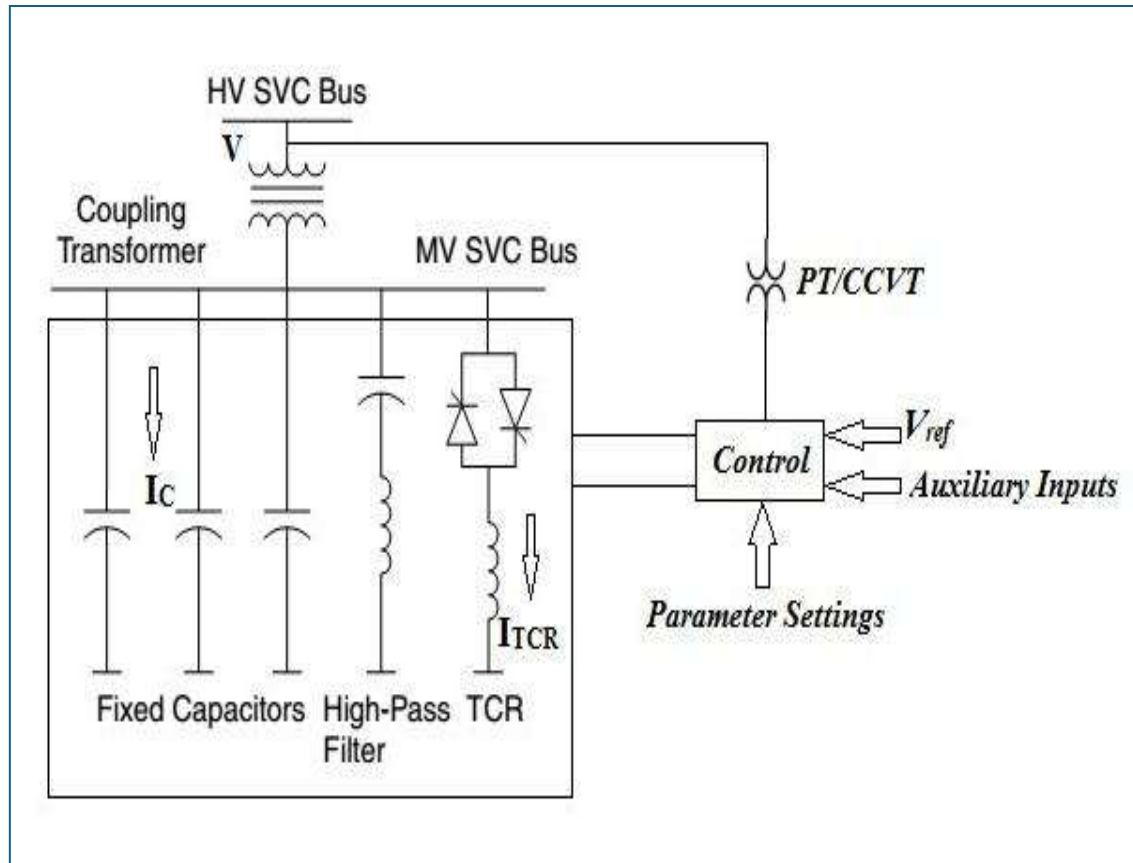


Fig. 2.9: A typical TCR-FC type SVC Configuration

In TSSC the capacitors are connected in series the thyristor of appropriate rating is connected in parallel of these capacitors and worked as bypass path as shown in figure 2.10. The stepwise reactive compensation is provided and controlled

by varying the number of the cascaded capacitors this thyristor-based valve is turn off to insert the compensation capacitor in a transmission line. The TSSC can provide the voltage as well as impedance compensation.

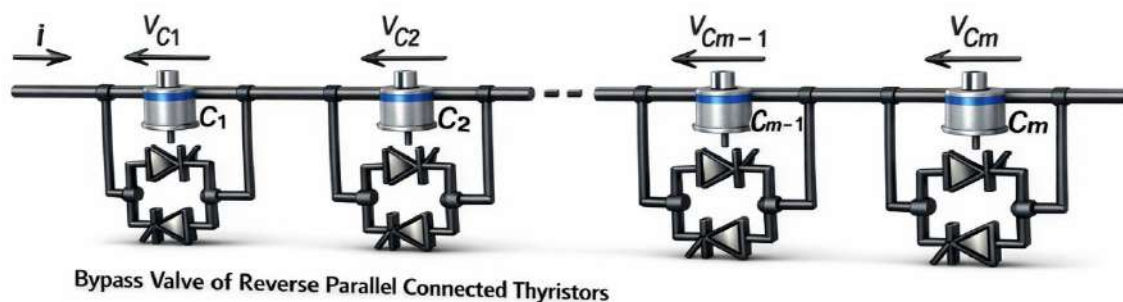


Fig. 2.10: TSSC typical Simple model

To discuss the voltage and current relation of TSSC the four modules are taken as shown in figure 2.11. For the purpose of compensation, the reactance of the module is set to compensation voltage $V_{C, max} = 4X_C I_{min}$ is the total effective

compensation voltage. The line current is used to turn off or on the bypass thyristor valve when the I_{min} is increased and it approaches I_{max} the capacitor is added in the system which results in a reduction of the system reactance.

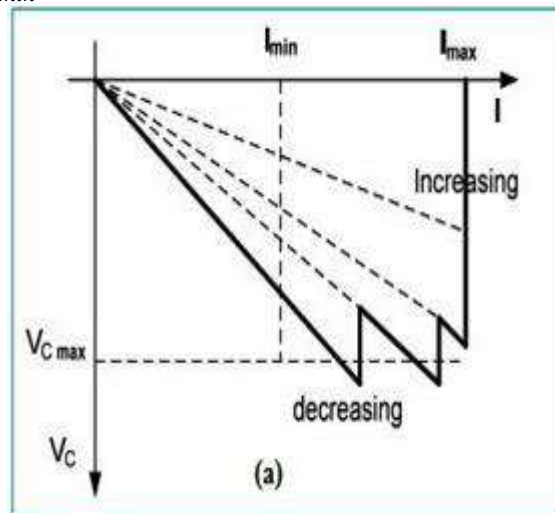


Fig. 2.11: Voltage Compensating Mode of TSSC.

The reactance mode is shown in Figure 2.12. Actually, the TSSC keeps the system reactance to

a maximum value at all operating value by line maximum current.

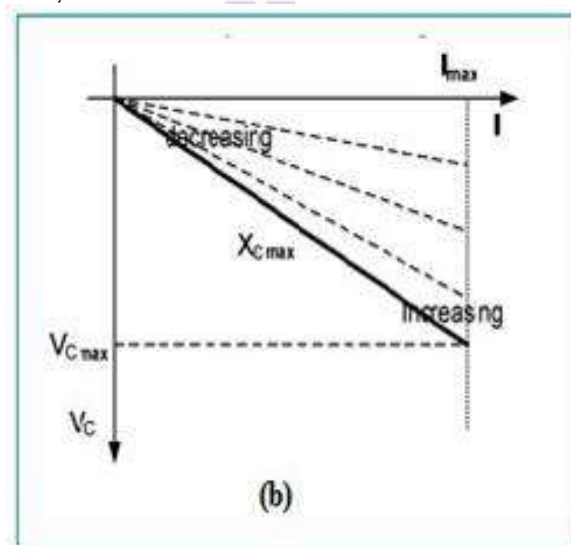


Fig.2.12: Reactance Control Mode

Reactive power is controlled synchronously using thyristor-based FACTS devices. System reactive impedance is controlled discretely or continuously, and compensation is achieved via voltage- or current-based converter devices. Synchronous converters are cascaded to form the SSSC, a voltage-based FACTS device. SSSC

operation depends on voltage angle variation, where the phase angle between current and voltage governs reactive power flow. Varying this angle controls reactive power in or out of the system for reactive and active power exchange, as shown in Figure 2.15.

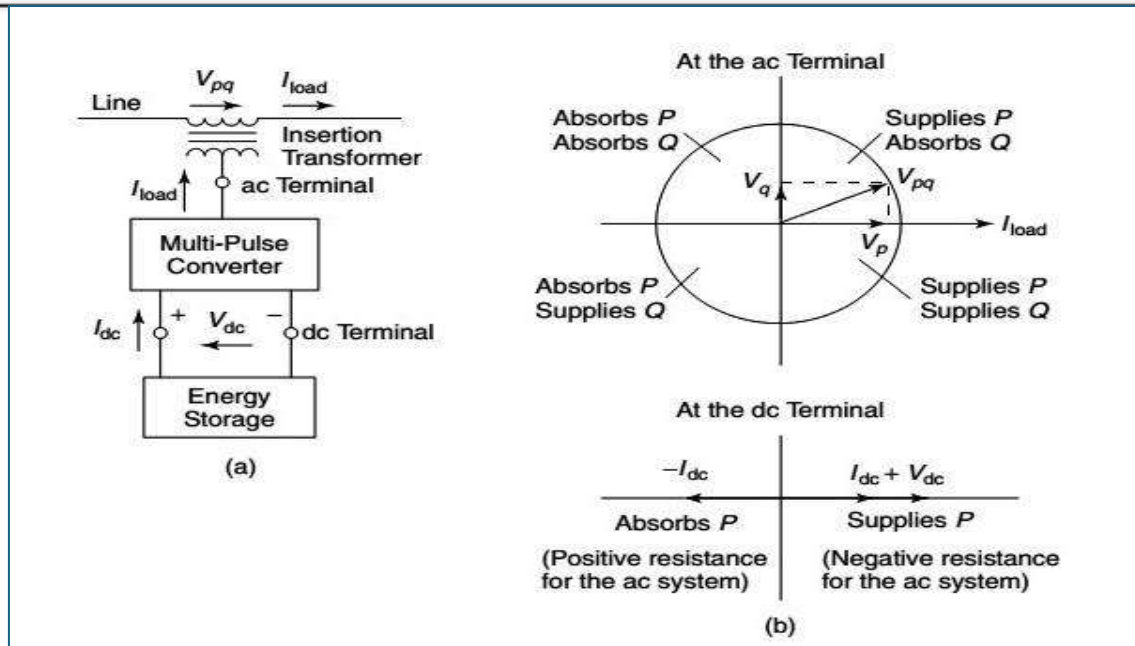


Fig. 2.15 (a) Basic Arrangement of SSSC, (b) Operating Modes of SSSC

A simple system is considered to discuss the operation of the SSSC, voltage V_s is sending voltage and V_R and V_L voltage is being dropped in a transmission line and V_C is capacitor voltage in figure 2.16. The voltage is in lagging quadrant

to decompensate the system inductance; actually, line inductance is in leading quadrature with the current so this voltage compensates the inductance.

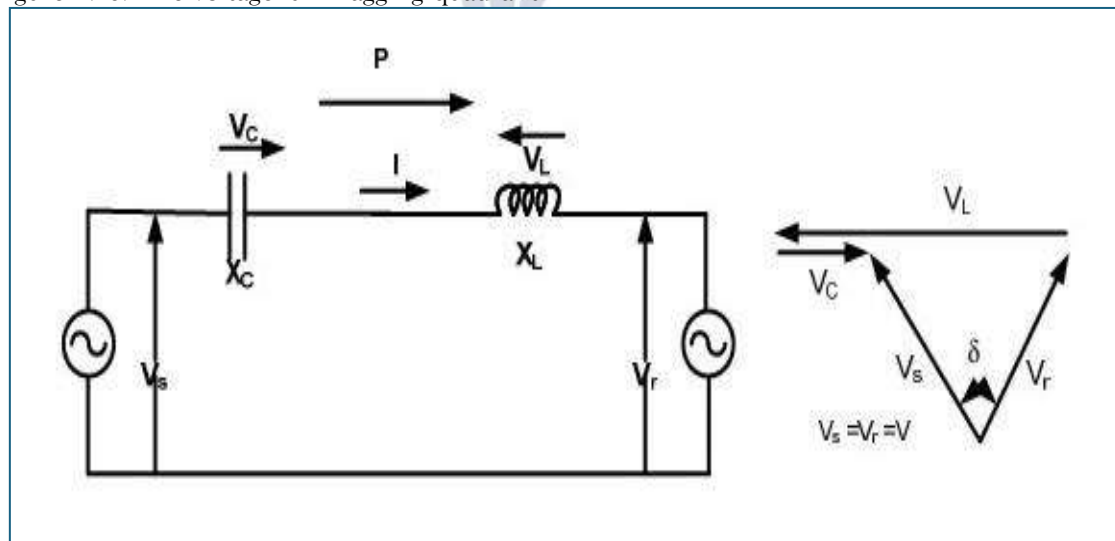


Fig. 2.16. A Two Machine system explaining SSSC Principle

STATCOM is a shunt-connected solid-state converter used to generate or absorb reactive power and control system parameters. An input energy storage element handles the required reactive power. Fed from a DC capacitor, STATCOM generates AC voltage in phase with the system [27]. Reactive power flows from higher

to lower magnitude and from leading to lagging sources. It monitors bus voltage, adjusts phase angle, and provides continuous reactive power without rotating components [28]. The configuration and equivalent circuit are shown in Figure 2.17.

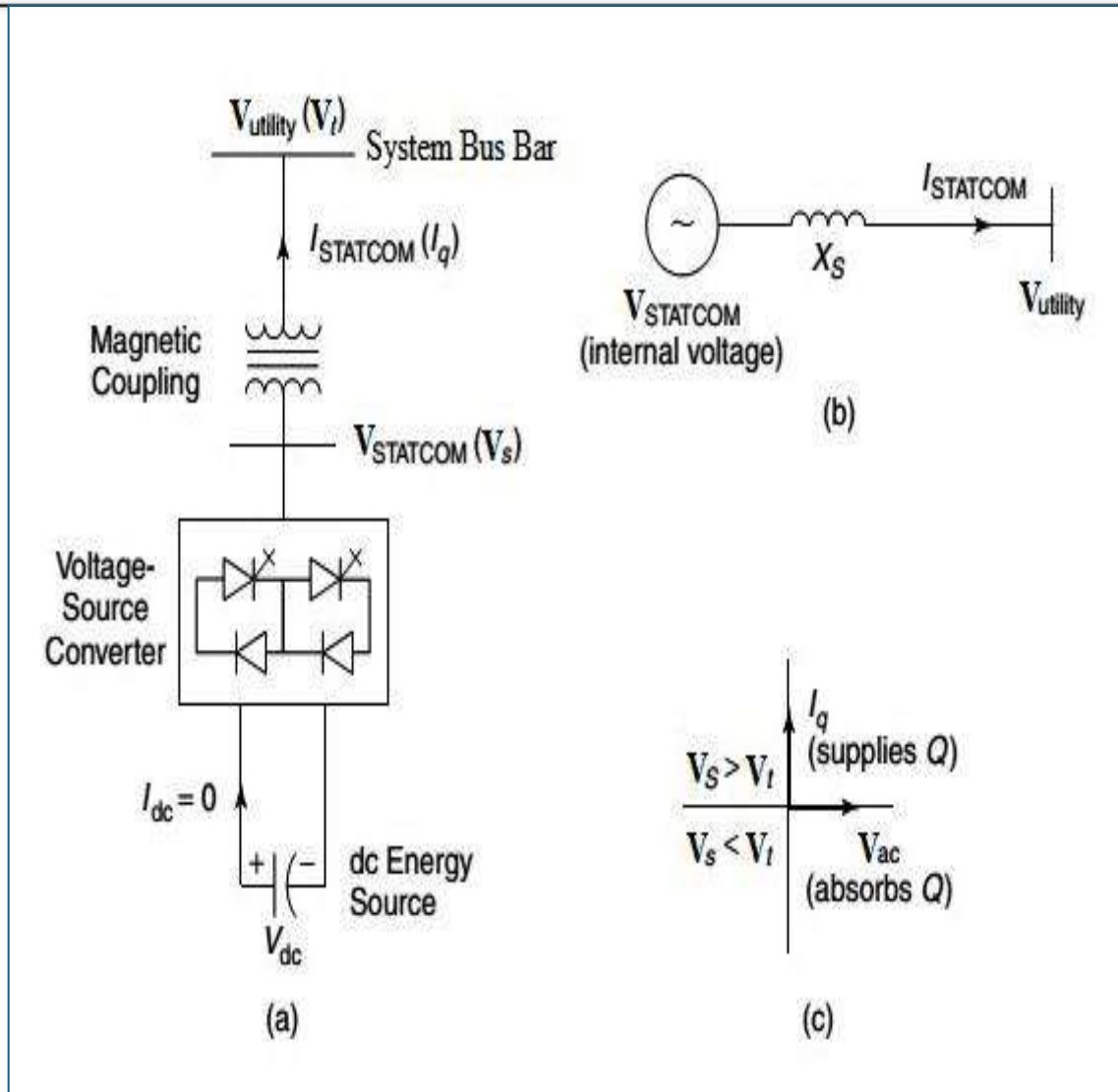


Fig. 2.17 STATCOM Configuration and Equivalent Circuit with Concept of Power Exchange

The STATCOM has a different mode of operation as:

- i) **Capacitive Mode:** in capacitive mode, the STATCOM is configured to produce the leading current that flows from the STATCOM to the connected system through the reactance Link, for this purpose the STATCOM voltage is kept high in magnitude than that of system bus voltage.
- ii) **Inductive Mode:** For the inductive mode the STATCOM is configured to produce the lagging current and the STATCOM voltage is produced in a lower value than that of bus voltage.
- iii) **Floating Mode:** To keep the STATCOM in floating mode, in which the STATCOM is connected with the system but no power flow is being occurred, the STATCOM is configured to achieve the same voltage as that of bus voltage.

Active and reactive powers are exchanged between STATCOM and the utility system. Power transfer is shown in Figure 2.17.

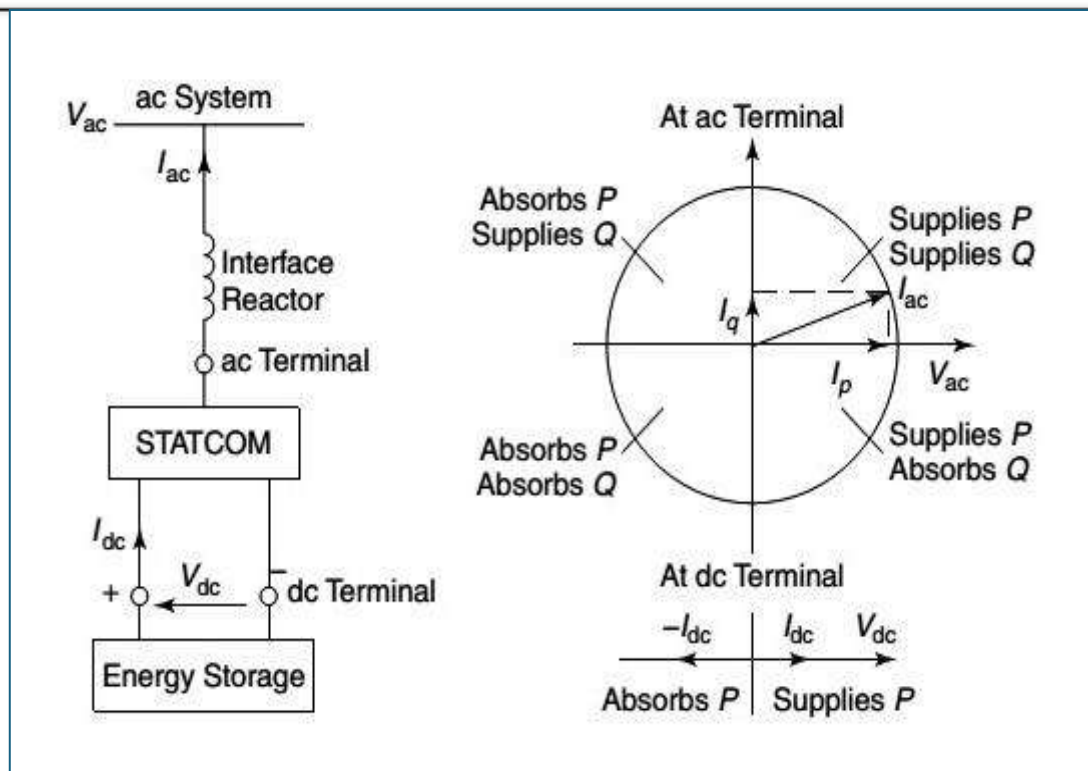


Fig. 2.18: Power exchange b/w STATCOM and Utility AC system.

Due to many factors, the STATCOM is more efficient and preferable option over other FACTS like SVC, the distinguishing feature make it more demanding are as;

STATCOM has quick dynamic response compared to other SVC

- Quicker dynamic reaction to faults than other compensators.
- STATCOM is capable of providing reactive as well as active power.
- It has low harmonic contents in output current.

- In the case of swell in system voltage, the STATCOM can provide the maximum reactive power even at this low voltage.

- The STATCOM can maintain its output current over leading and lagging maximum range independent of the Bus voltage.

- STATCOM provides the controllable voltage source connected in parallel to the system.

- During the operation, the STATCOM does not produce inrush current.

The performance of different FACTS is shown in the following Table 2.1.

ATTRIBUTES	SYNCHRONOUS CONDENSER	STATIC COMPENSATOR (e.g SVC)		SELF COMMUTATED COMPENSATOR (e.g STATCOM)
		TCR (with shunt capacitors if necessary)	TSC (with TCR if necessary)	
ACCURACY OF COMPENSATION	Good	Very Good	Good, very good with TCR	Excellent
CONTROL FLEXIBILITY	Good	Very Good	Good, very good with TCR	Excellent
REACTIVE POWER CAPABILITY	Leading/Lagging	Lagging /Leading indirect	Leading / Lagging indirect	Leading/Lagging
CONTROL	Continuous	Continuous	Discontinuous (cont. with TCR)	Continuous
RESPONSE TIME	Slow	Fast, 2 to 3 cycles	Fast, 2 to 3 cycles	Very fast (1 to 2 cycles) but depends on the control system and switching frequency
HARMONICS	Very High	Very high (large size filters are needed)	High, filters are necessary with TCR	Low, but depends on switching pattern
LOSSES	Moderate	Fair, but increase in lagging mode	Fair, but increases in leading mode	Low, but increases with switching frequency
PHASE BALANCING ABILITY	Limited	Good	Limited	Very good with 1- ϕ units, limited with 3- ϕ units
COST	High	Moderate	Moderate	Low to moderate

Table 2.1. Comparison of FACTS

3. Transmission line protection and impact of STATCOM

3.1. Introduction

For better and continuous power transmission, transmission lines and the grid have been enhanced. Transmission lines are vital assets and pass-through different areas facing harsh environments, making faults highly probable.

3.2. Transmission Line Protection Relays

3.2.1 Distance Relay

The distance relay protects transmission lines based on specific reactance and resistance per kilometer. Total line impedance depends on length, and the relay measures impedance to determine relay reach. In Figure 3.1, during a fault, voltage decreases and current increases, lowering impedance. If below reach, the relay isolates the faulty section.

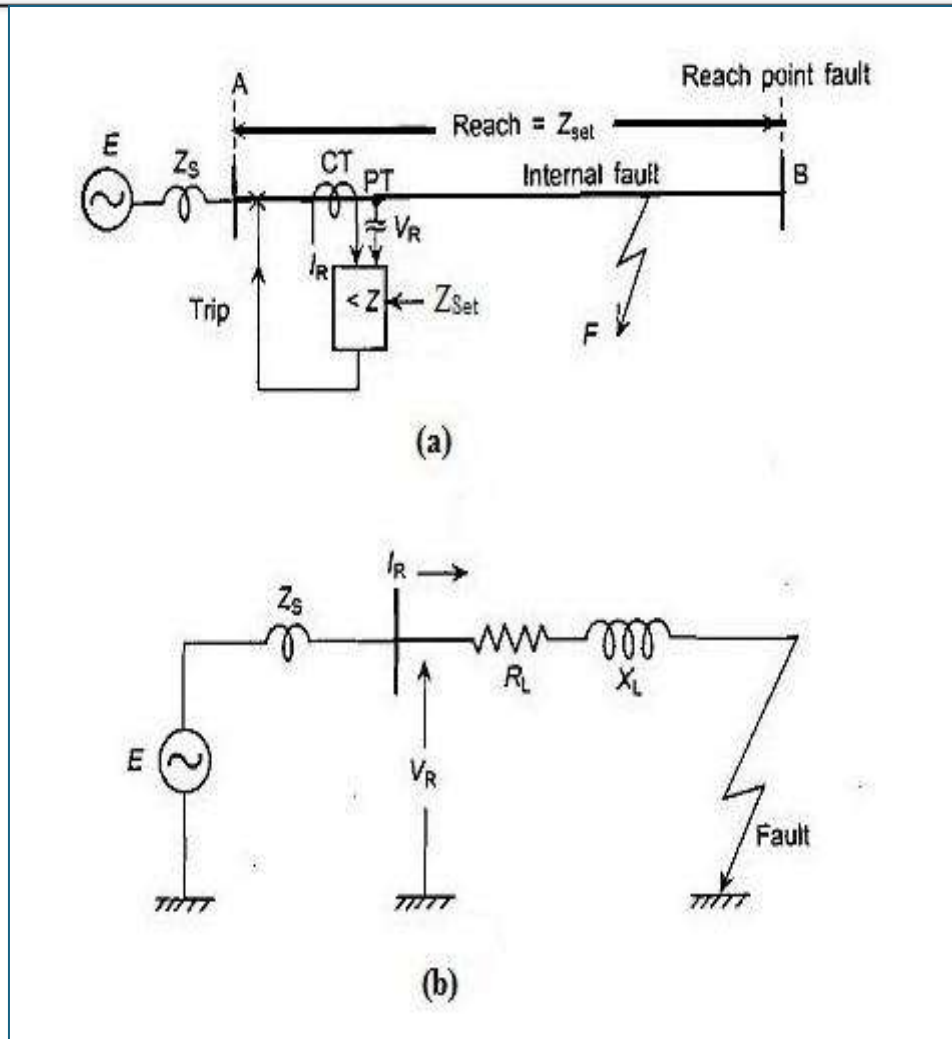


Fig. 3.1. Distance Relay Scheme

The relay has a predefined impedance and compares voltage and current during a fault. Current torque acts as actuating force, voltage torque as restraining force, and both are equal under normal conditions.

The torque produced by voltage coil will be equal to the torque applied by the current coil.

So,

$$T(V) = T(I) \quad (3.1)$$

$$T(V) \propto V^2 \quad (3.2)$$

$$T(I) \propto I^2 \quad (3.3)$$

$$T(V) = K_1 V^2 \quad (3.4)$$

$$T(I) = K_2 I^2 \quad (3.5)$$

As in normal condition, both the torque is equal so from eq 3.4 and 3.5

$$K_1 V^2 = K_2 I^2$$

$$(V/I)^2 = K_2/K_1$$

So impedance Z is given as;

$$Z^2 = K_2/K_1$$

$$\sqrt{R^2 + X^2}^2 = K_2/K_1 \quad (3.6)$$

This equation can be written in form of circle equation as

$$(R - 0)^2 + (X - 0)^2 = ((K_2/K_1)^{1/2})^2 \quad (3.7)$$

It is clear from the equation 3.7, that it is actually the equation of the circle passing through the center of the circle and its radius is $(K_2/K_1)^{1/2}$. It can be expressed in form of resistance and reactance as in Figure 3.2,

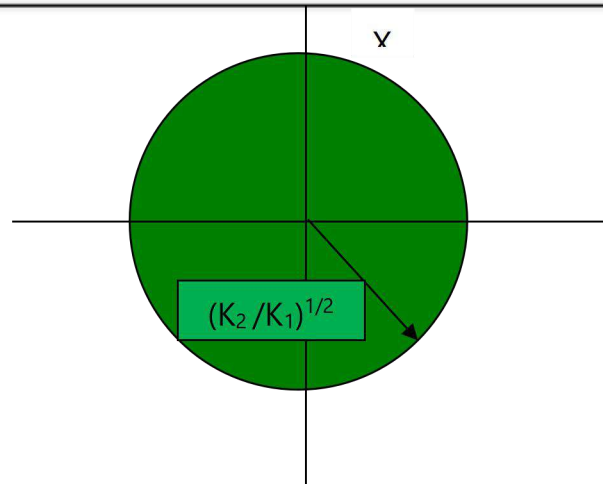


Fig. 3.2. Resistance and reactance

As mentioned above that the current torque is the Radius of the circle decreases and caused the actuating force so in a case of high value for K_1 relay to operate.

$$\text{If } \frac{|V_R|}{|I_R|} < |Z_{Set}| \text{ then Relay Sends Trip Command}$$

In other words, the relay will compare the line impedance with the preset impedance and can be expressed as;

$$\text{If } |Z_R| < |Z_{Set}| \text{ then relay sends trip command}$$

3.3. Distance Relay Characteristics

The distance relay characteristics are used to achieve the desired goal in view the beset protection features such as selectivity, discrimination, directional and non-directional tripping and relay reach boundary. To elaborate these characteristics the impedance is drawn on the R-X diagram, where R is along X-axis and Reactance X is along Y-axis. Depending on the

tripping zone these characteristics are designed. These different characteristics are as;

3.3.1 Plain Impedance Relay

The plain impedance relay is the under-impedance relay that will work on all angle, the relay will trip whenever the impedance value is less than preset impedance value. Relay location is considered at the center of a circle (0,0) and the tripping region is all around in 360° as shown in Figure 3.3.

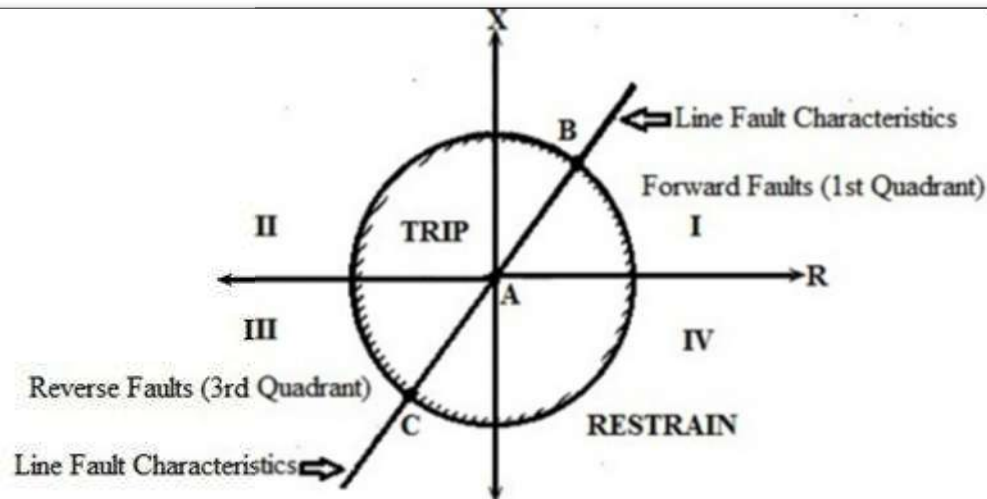


Figure 3.3. Plain Impedance Characteristics

3.3.2 Directional Distance Relay

In directional distance relay, the relay characteristics are designed to operate in a particular direction. As the impedance seen by the relay is given as,

$$Z_{Set} = K * \cos(\alpha - \lambda)$$

Where

K is constant,

α is line angle

and λ is maximum torque angle and also known as Relay Characteristics Angle RCA. In other words, the λ is the angle at which relay has maximum reach and this angle is set on a relay, α is the angle of a line during the fault. So, by

changing the angle α the impedance seen by the relay can be changed. If the angle λ and angle α same then the $\cos(\alpha - \lambda) = 1$, so the relay will have the maximum reach. It is clear that by change the value of RCA the relay can be made directional or the reach of the relay can be controlled and following are the different characteristics.

3.3.3 Resistance Relay

In this characteristic the relay only senses the resistive component of impedance and the both the angles are zero as shown in Figure 3.4, it is the example of directional distance relay that can sense the fault only along the X-axis.

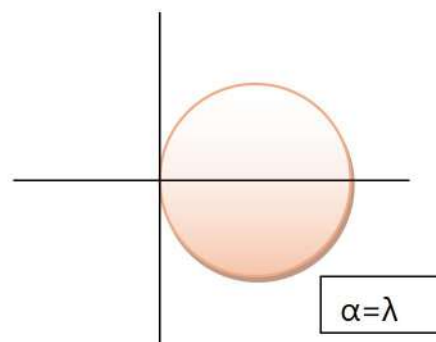


Figure 3.4. Resistance Distance Relay Characteristics

3.3.4 Reactance Impedance Distance Relay

This relay only senses the reactive component of impedance and operates only for this direction, the reactance value X_{Set} is preset on a relay, it is the directional form of plain impedance relay as

shown in figure 3.5. For ideal reactance distance relay, the both the angle is set to 90° and the relay only monitor the X value of the impedance and only trip along the Y-axis and all other points are restraint.

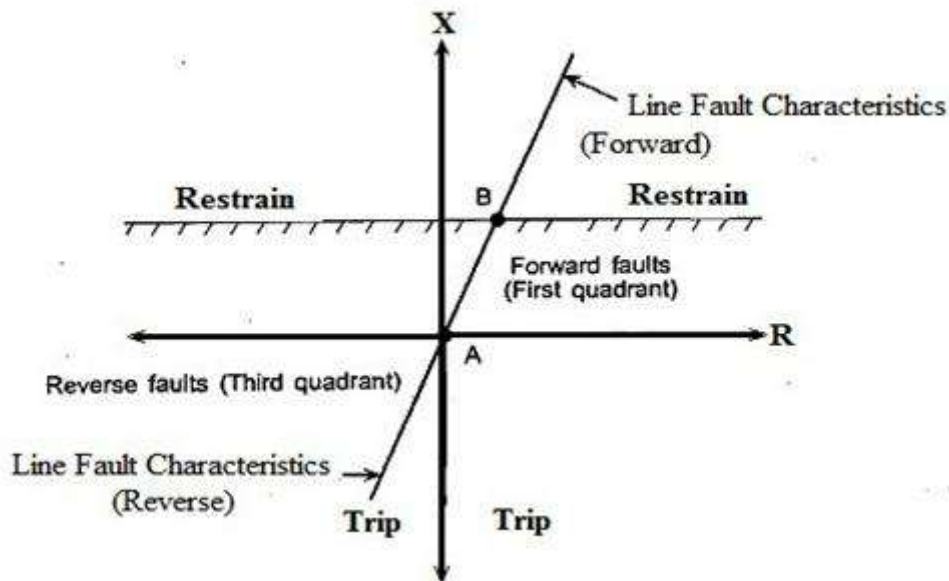


Figure 3.5. Reactance Distance Relay [26]

3.3.5 Mho Relay Characteristics

The mho characteristics relay is the directional version of the plain impedance relay, it is an

inherently directional relay, the center of the circle is shifted and it is also in the first quadrant as shown in Figure 3.6.

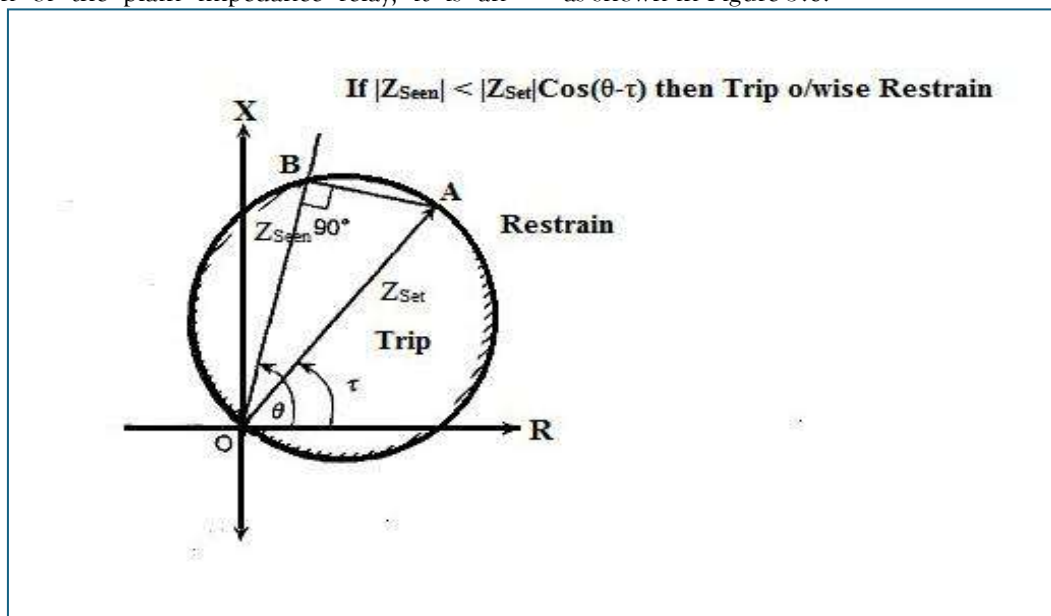


Figure 3.6. Mho Characteristics of Distance relay.

The mho characteristic is used on the heavily loaded lines to overcome the problem of load encroachment and has the great stability under normal as well as power swing in the system.

3.3.6 Quadrilateral Impedance Relay

The quadrilateral characteristics monitor the resistive and reactive component of the

transmission line, in modern relays, this feature is used and it is very efficient in case of short line or in the fault which has high arc resistance. R-X diagram for the quadrilateral relay is shown in Figure 3.7.

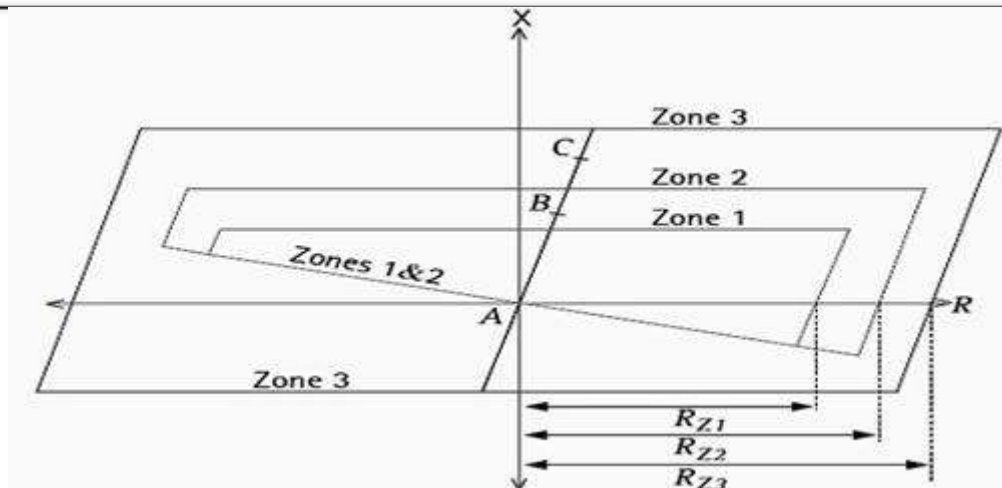


Figure 3.7. Quadrilateral impedance Relay R-X diagram.

3.4 Protection Zones

The line to be protected by the distance relay is divided into different zones. Each zone has preset Reach and time setting, Zone-1 is usually 80-85 % of total line length and its tripping time is instantaneous, Zone-2 is 120% of protected

transmission line and having tripping time 300-450 m sec, Zone-3 full line length plus the 50% of next transmission line with tripping time of 700-1000 m sec, the Zone-3 also works as a backup protection for the next line. The general 3 zones impedance relay is shown in figure 3.8.

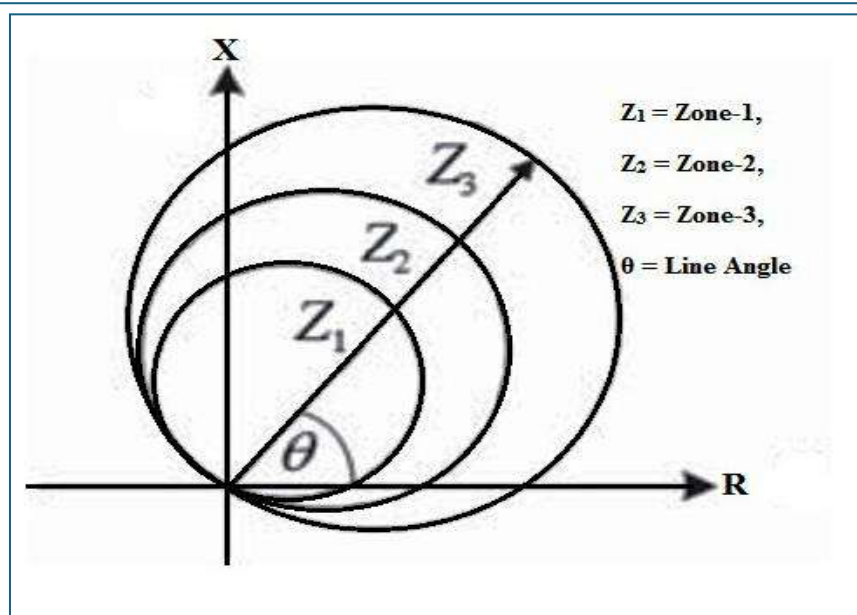
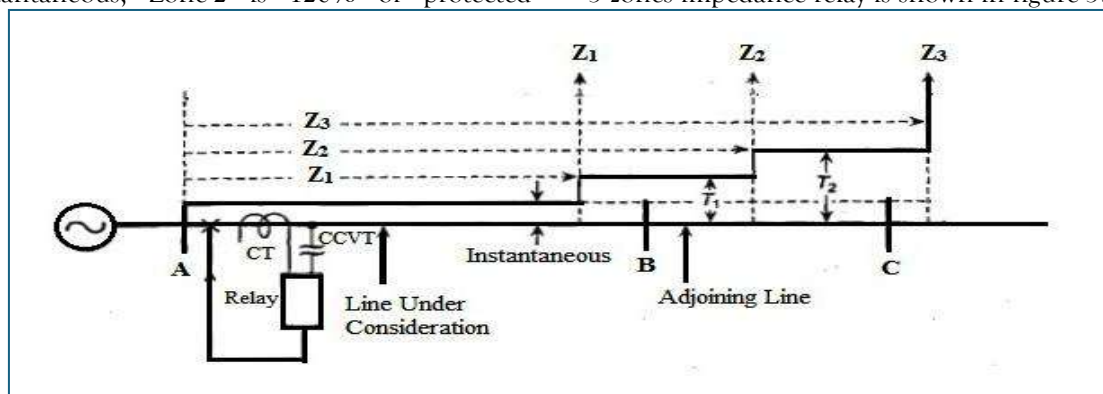


Figure 3.8. 3 Zones of impedance relay.

3.5. FACTS devices impact on Distance Relay

An effective protection scheme must fulfill attributes such as reliability, selectivity, dependability, sensitivity, and fast fault clearance. The introduction of FACTS devices affects system frequency, current, and voltage, impacting these attributes. STATCOM can cause rapid changes in line impedance and power angle. As distance relays calculate voltage-current ratios, STATCOM injection of current and voltage for reactive power compensation may lead to mal-operation.

In this work the impact of the STATCOM is analyzed simulating the simple power system with Distance relay.

4. Modeling of system under study

This chapter considers the impact of STATCOM on the distance relay. A high-voltage transmission line with actual parameters and D60 relay settings is simulated. The 220KV KAPCO-Multan transmission line, part of Pakistan's interconnected 220KV and 500KV AC system operated by NTDCL, is studied. A 100MVAR STATCOM is proposed to reduce transmission losses and maintain voltage. Before implementation, its impact on protection relays is analyzed. STATCOM and distance relay are modeled and verified in MATLAB/SIMULINK using real-time field parameters and settings for practical analysis.

4.1. EHV Transmission Line under study

EHV transmission line 220KV KAPCO-Multan is taken for study, the actual parameters of line and relay settings are modeled in MATLAB.

Transmission Line Data is given as;

Total Length of the T/L to be protected. AB (KAPCO-Multan T/L) = 100 km;

Transmission Line Conductor: ZEBRA (ACSR)

Max. Current Carrying Capacity: 810 Amps

Primary Rated Voltage: 220kV

Secondary Rated Voltage: 110V

Primary CT current: 1200A

Secondary CT current: 5A

M-Factor = CCVT Ratio / CT Ratio = $(220\text{kV}/110\text{V}) / (1200\text{A}/5\text{A}) = 8.3$

Positive Sequence Impedance = $Z_1 = (0.06 + j0.392) \Omega/\text{km}$

Positive Sequence Impedance = $Z_1 = 0.3967 / 81.3^\circ \Omega/\text{km}$

Positive Sequence Resistance = $R_1 = 0.060 \Omega/\text{km}$

Positive Sequence Inductance = $L_1 = 1.28 \text{ mH}/\text{km}$

Positive Sequence Capacitance = $C_1 = 13.74 \text{ nF}/\text{km}$

Zero Sequence Impedance = $Z_0 = (0.246 + j1.1248) \Omega/\text{km}$

Zero Sequence Impedance = $Z_0 = 1.1514 / 77.66^\circ \Omega/\text{km}$

Zero Sequence resistance = $R_0 = 0.246 \Omega/\text{km}$

Zero sequence Inductance = $L_0 = 3.709 \text{ mH}/\text{km}$

Zero Sequence Capacitance = $C_0 = 7.951 \text{ nF}/\text{km}$

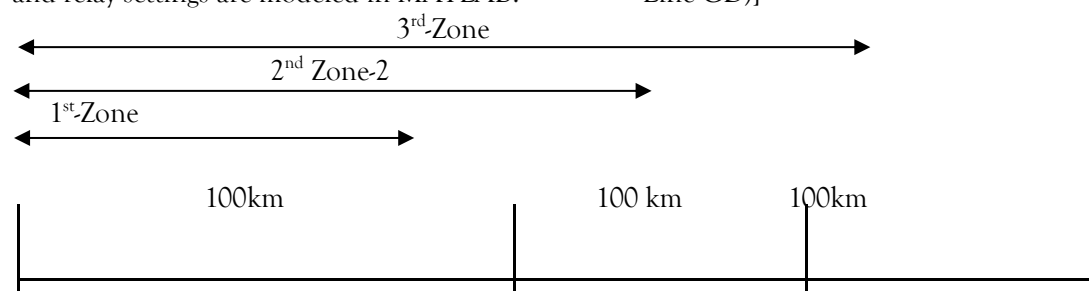
As first zone of the distance relay is 80-85% of total line length, so 1st zone;

Zone-1 = Protects 85% of the Line AB.

And 2nd zone is given as,

Zone-2 = Protects [Line AB + (50% of Line BC)].

Zone-3 = Protects [Line AB + Line BC + (25% of Line CD)]



A

4.2 IMPEDANCE Relay Settings**4.2.1 Phase-to-Phase Reach Setting**

For phase-to-phase impedance reach setting Positive sequence, Impedance is taken for Reach settings.

- 1st ZONE SETTINGS, Z_1**

1st Zone length = L_1 = 85% o Line Length
Multan-KAPCO = $85/100 \times 100 = 85$ km

Impedance Setting of $Z_1 = Z_{1-P(ph-ph)} = L_{Z1} \times 0.396$

Impedance Setting of $Z_1 = Z_{1-P(ph-ph)} = 85 \times 0.396$
 $\Omega/\text{km} = 33.66 \underline{\Omega}$

$Z_{1-P(ph-ph)} = 33.66 \underline{\Omega} = 33.66/81.3^\circ \underline{\Omega}$

Impedance Setting of Z_1 (Secondary value) =

$Z_{1-S(ph-ph)} = 33.66 \underline{\Omega} / 8 = 4.0554 \underline{\Omega}$

$Z_{1-S(ph-ph)} = 4.0554 \underline{\Omega} = 4.0554 / 81.3^\circ \underline{\Omega}$

- 2nd ZONE SETTINGS, Z_2**

Length for 2nd Zone = L_2 = Length of Line
Multan-KAPCO + 50% of Next shortest line

= $100\text{km} + (50\% \text{ of } 60\text{km}) = 100\text{km} + 30\text{km} =$
 130km

B

C

D

Impedance Setting of $Z_2 (= Z_{2-P(ph-ph)} = L_{Z2} \times 0.396$
= $130\text{km} \times 0.396$

Impedance Setting of $Z_2 = Z_{2-P(ph-ph)} = 51.48 \underline{\Omega}$

$Z_{2-P(ph-ph)} = 51.48 \underline{\Omega} = 51.48/81.3^\circ \underline{\Omega}$

Impedance Setting of Z_2 (Secondary value) = $Z_{2-S(ph-ph)}$

= $51.48 \underline{\Omega} / 8.3 = 6.2 \underline{\Omega}$

$Z_{2-S(ph-ph)} = 6.2 \underline{\Omega} = 6.2/81.3^\circ \underline{\Omega}$

- 3rd ZONE SETTINGS, Z_3**

Line Length for 3rd Zone = L_3 = Length of Line
Multan-KAPCO + Length of Line shortest+ 25%
of length of longest Line = $100\text{km} + 60\text{km} + (25\%$
of $60\text{km})$

= $100\text{km} + 60\text{km} + 15\text{km} = 175\text{km}$

3rd Zone Length = $L_3 = 175\text{km}$

Impedance Setting of $Z_3 = Z_{3-P(ph-ph)} = L_{Z3} \times 0.396$

= $175\text{km} \times 0.396$

Impedance Setting of $Z_3 = Z_{3-P(ph-ph)} = 69.3 \underline{\Omega}$

$Z_{3-P(ph-ph)} = 69.3 \underline{\Omega} = 69.3 / 81.3^\circ \underline{\Omega}$

Impedance Setting of Z_3 (Secondary value) = $Z_{3-S(ph-ph)}$

= $69.3 \underline{\Omega} / 8.3 = 8.35 \underline{\Omega}$

$Z_{3-S(ph-ph)} = 8.35 \underline{\Omega} = 8.35 / 81.3^\circ \underline{\Omega}$

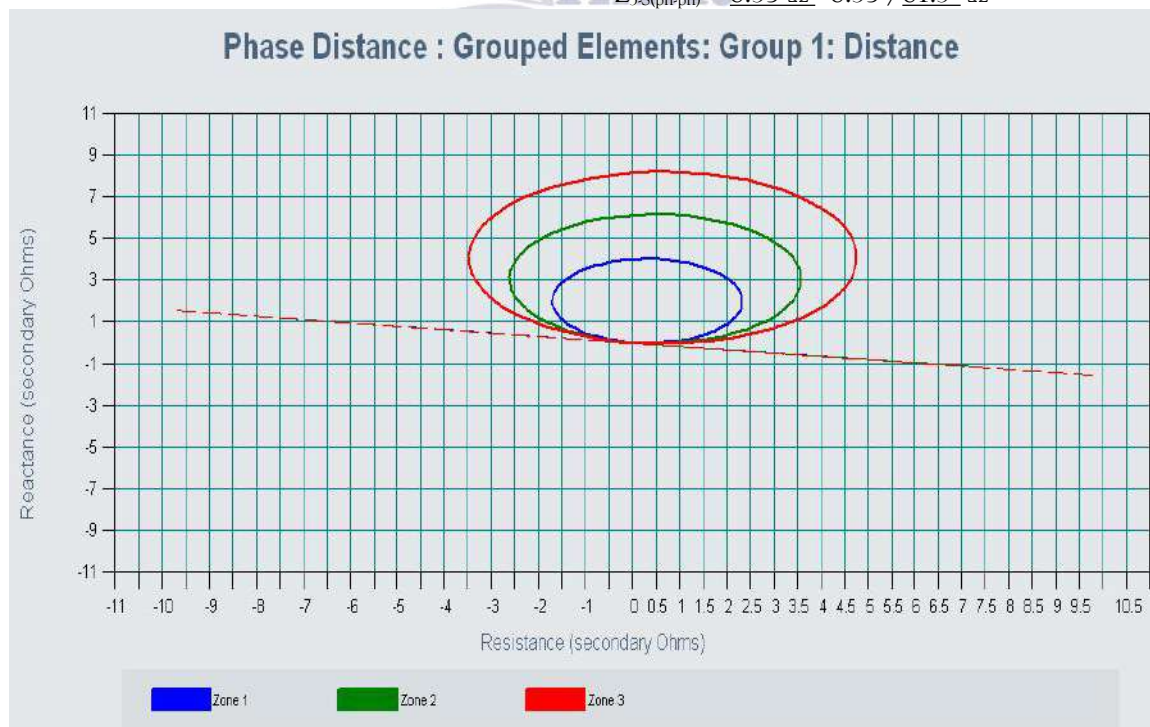


Figure 4.1. Zone Reach Settings for 3-phases & ph-ph Faults

4.2.1.1 Phase-to-Ground Fault Reach Setting

For phase-to-Ground impedance reach setting Negative sequence, Impedance is taken for Reach

settings.

- **1st ZONE SETTINGS, Z_1**

Length of $Z_1 = L_1 = 85\%$ of Line Length Multan-KAPCO = $85/100 \times 100 = 85$ km

Impedance Setting of $Z_1 = Z_{1-P(Ph-G)} = L_{Z1} \times 1.1514$

Impedance Setting of $Z_1 = Z_{1-P(Ph-G)} = 85 \text{ km} \times 1.1514 \Omega/\text{km} = 97.87 \Omega$

$Z_{1-P(Ph-G)} = 97.87 \Omega = 97.867 / 77.66^\circ \Omega$

Impedance Reach Setting of Z_1 (Secondary Value) = $Z_{1-S(Ph-G)} = 97.87 \Omega / 8.3 = 11.79 \Omega$

$Z_{1-S(Ph-G)} = 11.79 \Omega = 11.79 / 77.66^\circ \Omega$

- **2nd ZONE SETTINGS, Z_2**

Length for 2nd Zone = $L_2 =$ Length of Line Multan-KAPCO + 50% of Next shortest line = $100\text{km} + (50\% \text{ of } 60\text{km}) = 100\text{km} + 30\text{km} = 130\text{km}$

Impedance Reach Setting of $Z_2 = Z_{2-P(Ph-G)} = L_{Z2} \times 1.1514$

Impedance Reach Setting of $Z_2 = Z_{2-P(Ph-G)} = 130\text{km} \times 1.1514 \Omega/\text{km} = 149.68 \Omega_{Z_{2-P(ph-ph)}}$

$Z_{2-P(Ph-G)} = 149.68 \Omega = 149.68 / 77.66^\circ \Omega$

Impedance Setting of Z_2 (Secondary Value) = Z_2

$S_{(ph-ph)} = 149.68 \Omega / 8.3 = 18.1 \Omega$

$Z_{2-S(Ph-G)} = 18.1 \Omega = 18.1 / 77.66^\circ \Omega$

- **3rd ZONE SETTINGS, Z_3**

Line Length for 3rd Zone = $L_3 =$ Length of Line Multan-KAPCO + Length of Line shortest + 25% of length of longest Line = $100\text{km} + 60\text{km} + (25\% \text{ of } 60\text{km})$

= $100\text{km} + 60\text{km} + 15\text{km} = 175\text{km}$

3rd Zone Length = $L_3 = 175\text{km}$

Impedance Reach Setting of $Z_3 = Z_{3-P(Ph-G)} = L_{Z3} \times 1.1514 = 175\text{km} \times 1.1514 \Omega/\text{km} = 201.5 \Omega$

$Z_{3-P(Ph-G)} = 201.5 \Omega = 201.5 / 77.66^\circ \Omega$

Impedance Reach Setting of Z_3 (Secondary Value) = $Z_{3-S(Ph-G)} = 201.5 \Omega / 8.3 = 24.3 \Omega$

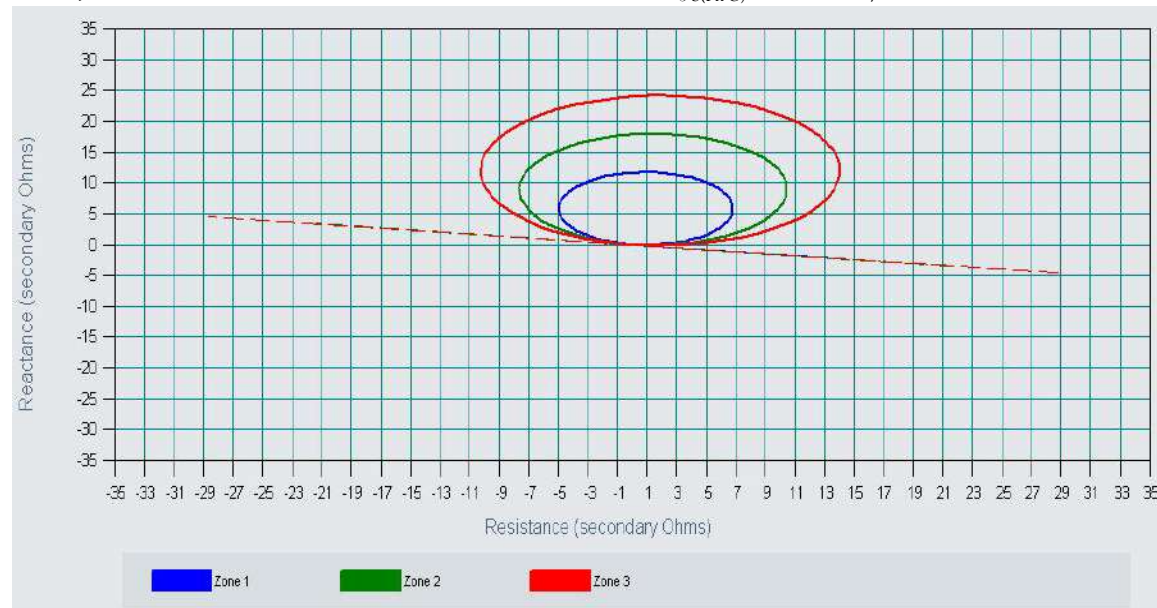


Figure 4.2. Zone Reach Settings for *ph-G* Faults

4.3 System Modeling in MATLAB/SIMULINK

The proposed model has two sections: the 220KV Multan-KAPCO transmission line with D60 distance relay for all fault types, including

impedance calculations and zone coordination, and a 100MVAR STATCOM. Faults are simulated with STATCOM connected at different locations. The block diagram of the model is shown in Figure 4.3.

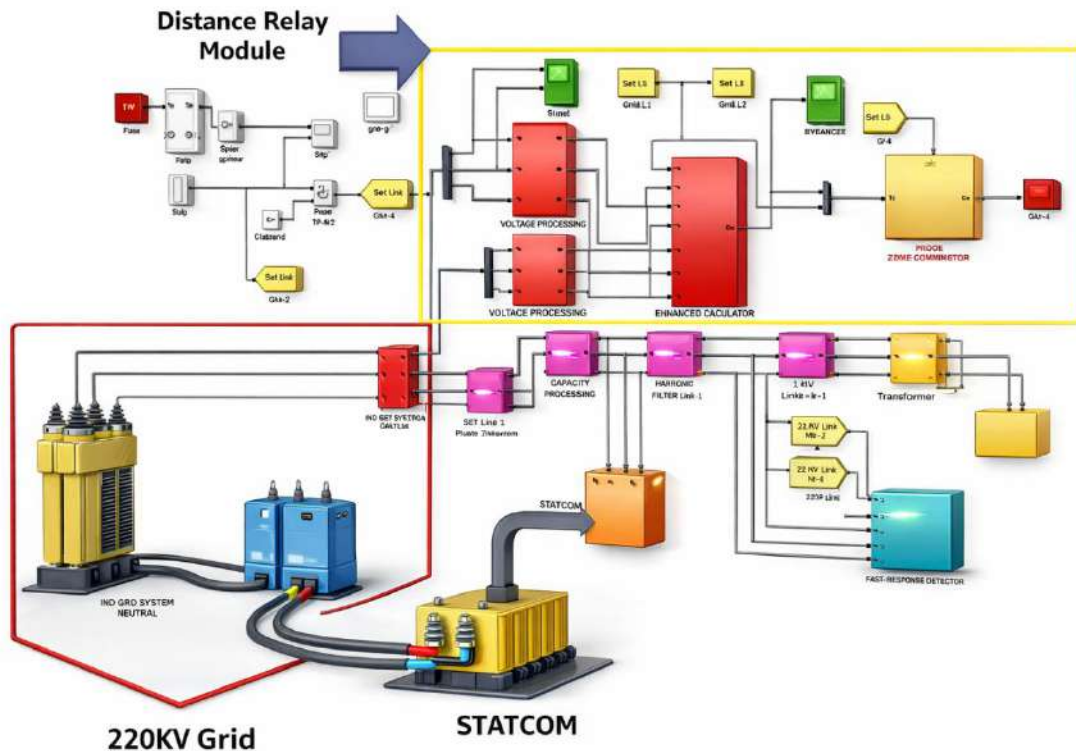


Figure 4.3. Block Diagram for the model.

The distance relay is simulated as shown in Figure 4.4.

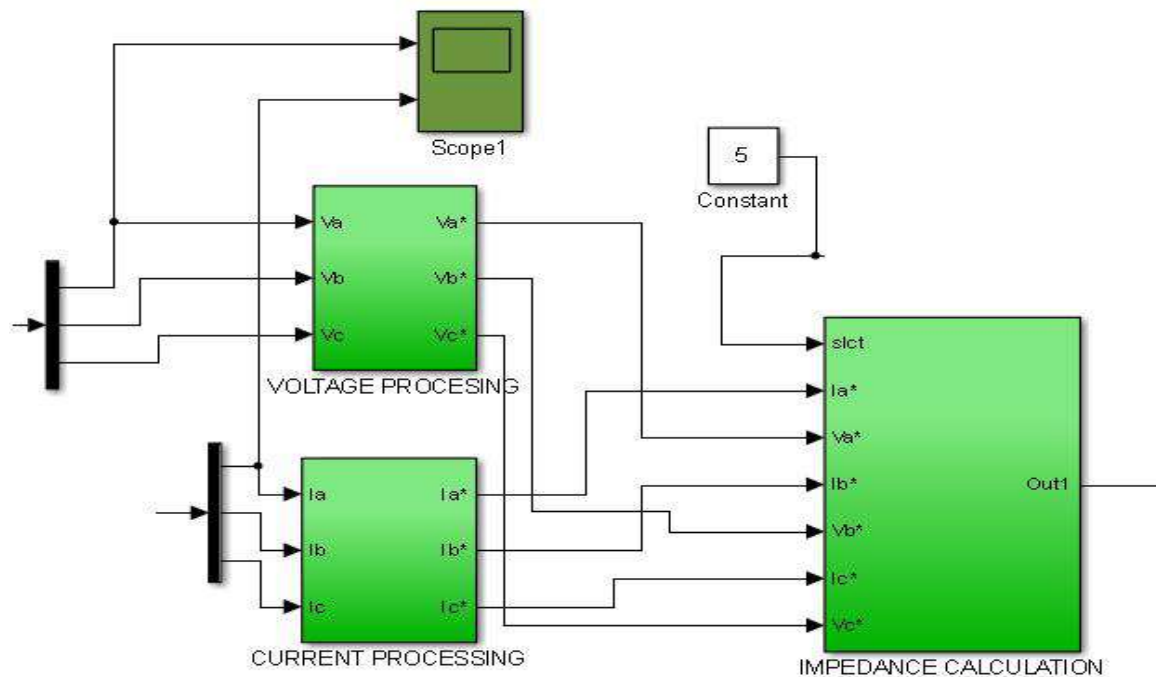


Figure 4.4. Distance relay Model

In distance relay, the major parts are current voltage processing and relay Zones coordination

the relay zones coordination is simulated as in Figure 4.5.

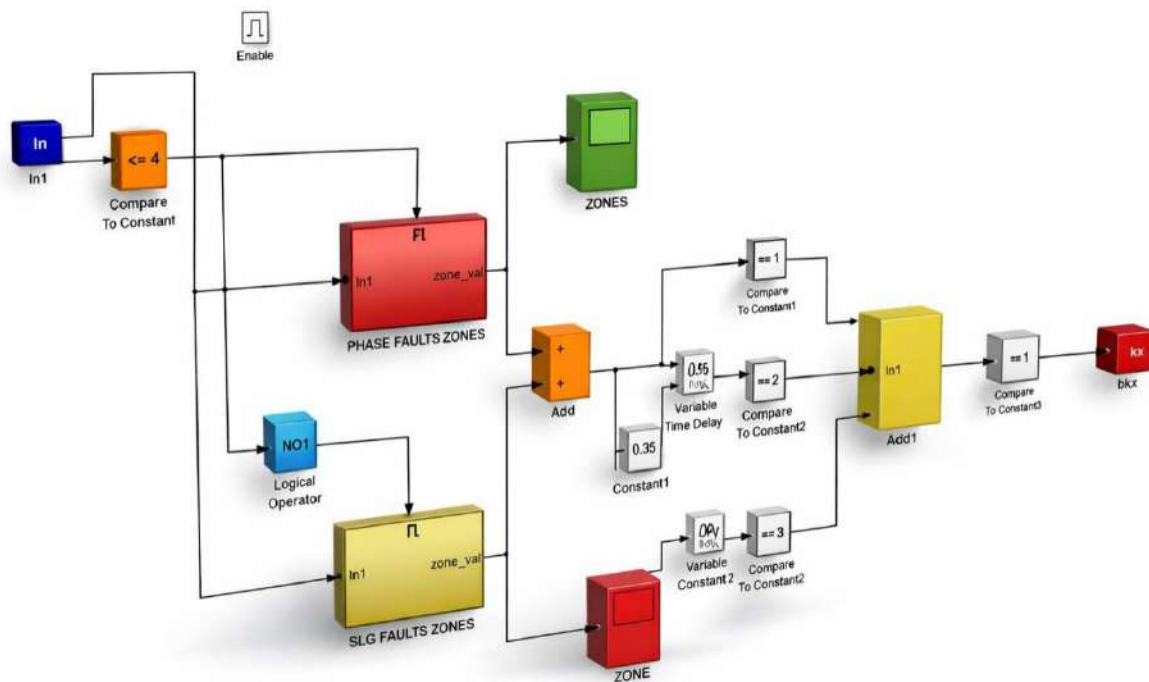


Figure 4.5. Distance relay Zones Coordination Model.

- STATCOM Model Simulation

The STATCOM for 220KV rated voltage with ± 100 MVAR STATCOM is simulated for proposed scheme and voltage is controlled by using this STATCOM.

The distinguishing parameter for STATCOM are given as,
 $C = 700 \text{ mF}$; $V_{dc} = 10 \text{ KV}$; $L = 0.65\text{e-}3 \text{ H}$; $R = 0.65\text{e-}3 \Omega$

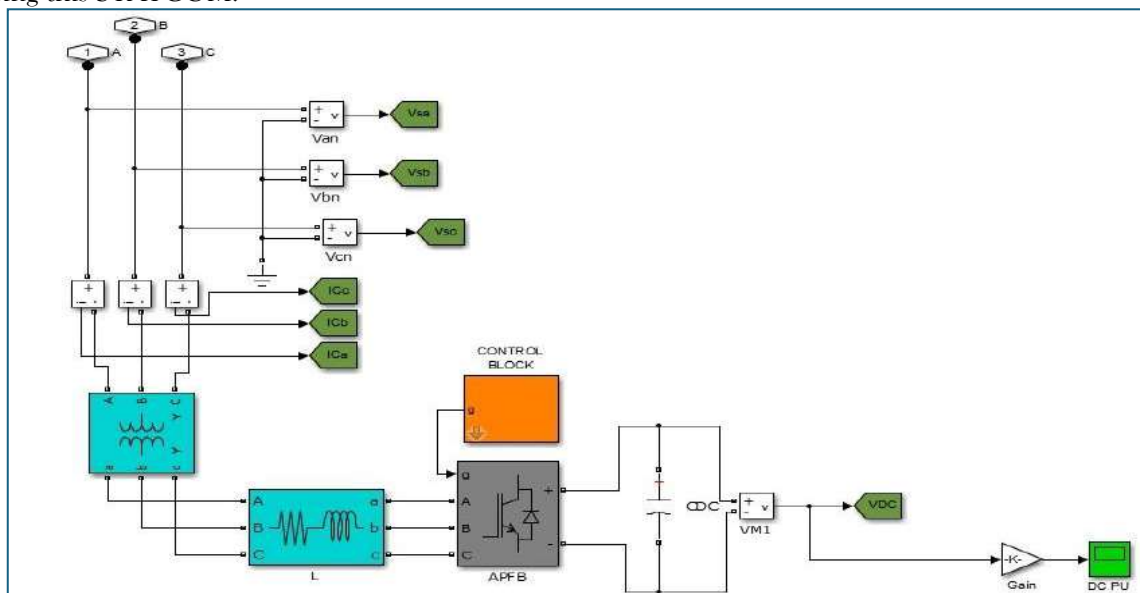


Figure 4.6. STATCOM Modeling.

The leakage inductor is denoted by L and compensation Capacitor is denoted by C_{dc} , the active components are used to implement the

control scheme. The control scheme produces the pulse and effectively controls the reactive currents I_{Ca} , I_{Cb} , and I_{Cc} .

5 Results and conclusions

5.1. 3-phase Fault Simulation Results

For analysis of the 3-phase fault the simulated model is considered the pre-fault conditions are as follows;

Line Voltage= 224KV

Line Current= 600AMPs

Relay Status= Not Operate

After 2 seconds of above pre-fault condition, the 3-Phase fault is simulated in absence of

STATCOM and results are analyzed and then the same fault is injected in presence of STATCOM installed at, 24, 60 and 78% of transmission line length. The system conditions in form of voltage and current are shown in figure 5.1.

The distance relay D60 reach settings are set as;

1st Zone Impedance Reach = 4.02 Ω

2nd Zone Impedance Reach = 6.2 Ω

3rd Zone Impedance Reach = 8.35 Ω



Figure 5.1. Pre-Fault and Post Fault System Voltage & Current.

5.2.1. Fault at 24% (24km) of Transmission Line Length

The fault at 24% of transmission line before the mid-point of the line at 24 KM with and without

of STATCOM is simulated. The impedance reach for the relay in both cases is shown in the Figure.5.2

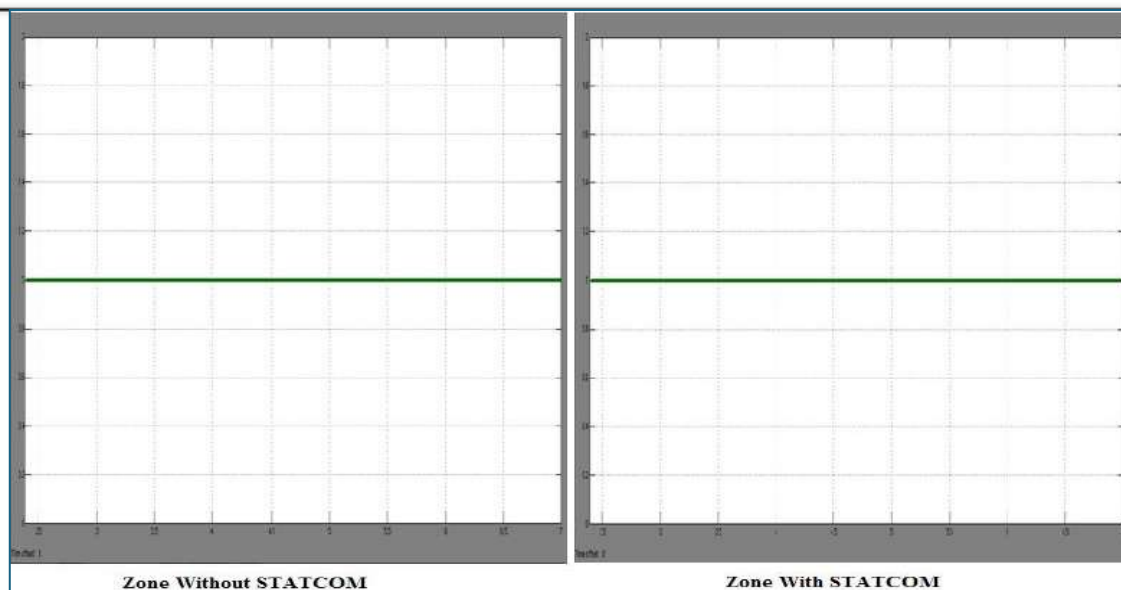


Figure 5.2. With and Without STATCOM Protection Zone Comparison at 24% Line Length.

Reach setting comparison with and without STATCOM at 24% of T/L length. After simulation, the same impedance relay was tested

and the reach graphs are extracted as in Figure 5.3

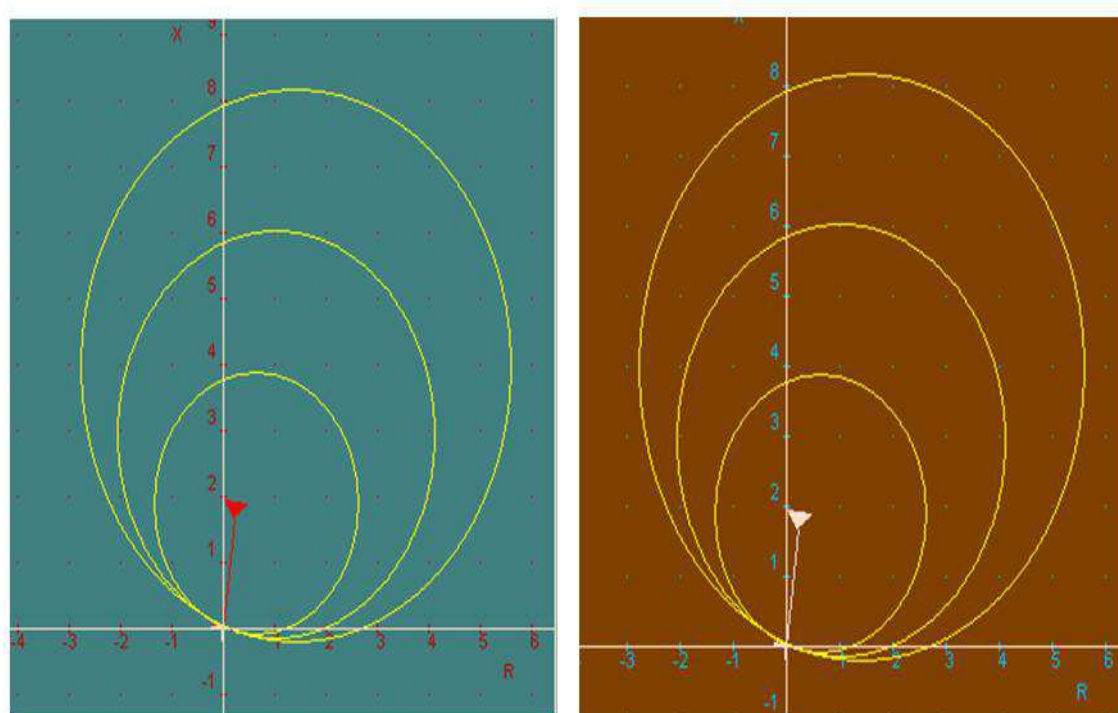


Figure 5.3. Impedance Reach 24% of line with & without STATCOM

- **Analysis & remarks**
After faulting at 24% of transmission line length, these prominent factors are observed as follow,
The impedance is not affected by the STATCOM and it is same as shown in Figure 5.2.

The relay operates in the same zone and the tripping time is 20 msec.

The same results are extracted from the FREJA testing software.

5.2.2. Fault at 60% (60km) of Transmission Line Length

The fault at 60% of transmission line before the mid-point of the line at 60 KM with and without of STATCOM is simulated. The impedance

reach for the relay in both cases is shown in the Figure.5.4.

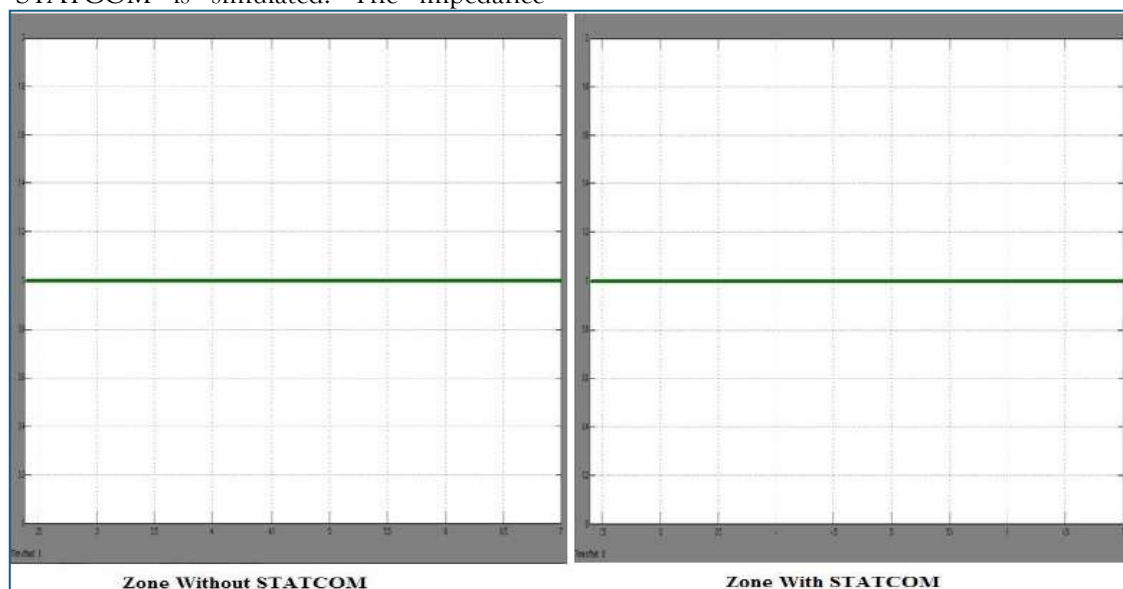


Figure 5.4. With and Without STATCOM Protection Zone Comparison at 60% Line Length.

Reach setting comparison with and without STATCOM at 60% of T/L length. After simulation, the same impedance relay was tested

and the reach graphs are extracted as in Figure 5.5.

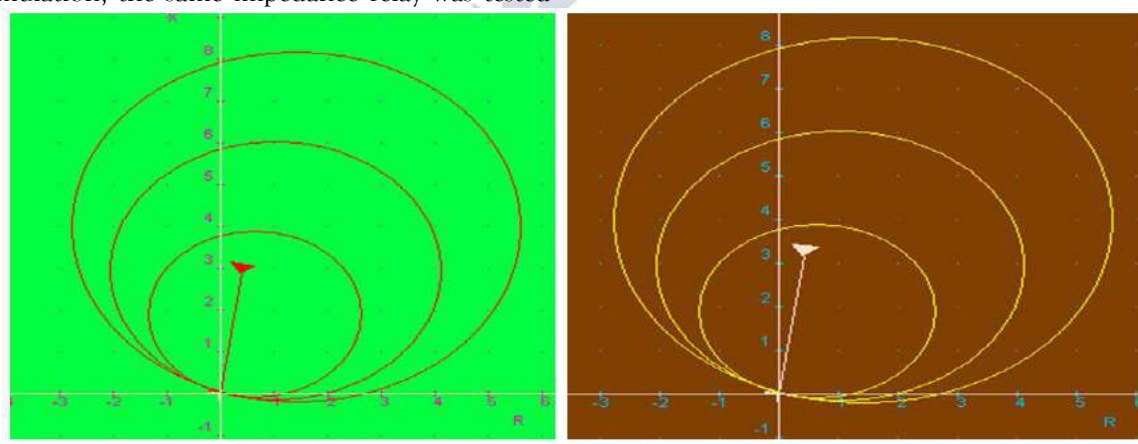


Fig. 5.5 Impedance Reach 60% of line with & without STATCOM

- Analysis & remarks**

After faulting at 60% of transmission line length these prominent factors are observed as follow,
From Simulation results, it is evident that at 60% of T/L distance (60km) i.e after Midpoint;
Impedance without STATCOM is = 3Ω and
Impedance with STATCOM is = 3.2Ω .
%age Increase in Impedance = 6.7%
Tripping Zones with & without STATCOM
Reach with & w/o STATCOM.

The impedance is affected by injection of STATCOM in system,

So, it can be concluded that STATCOM inside the fault loop will affect the reach of reach.

5.2.3. Fault at 78% of Transmission Line Length

The fault at 78% of transmission line before the mid-point of the line at 78 KM with and without of STATCOM is simulated. The impedance reach for the relay in both cases is shown in the Figure.5.6.

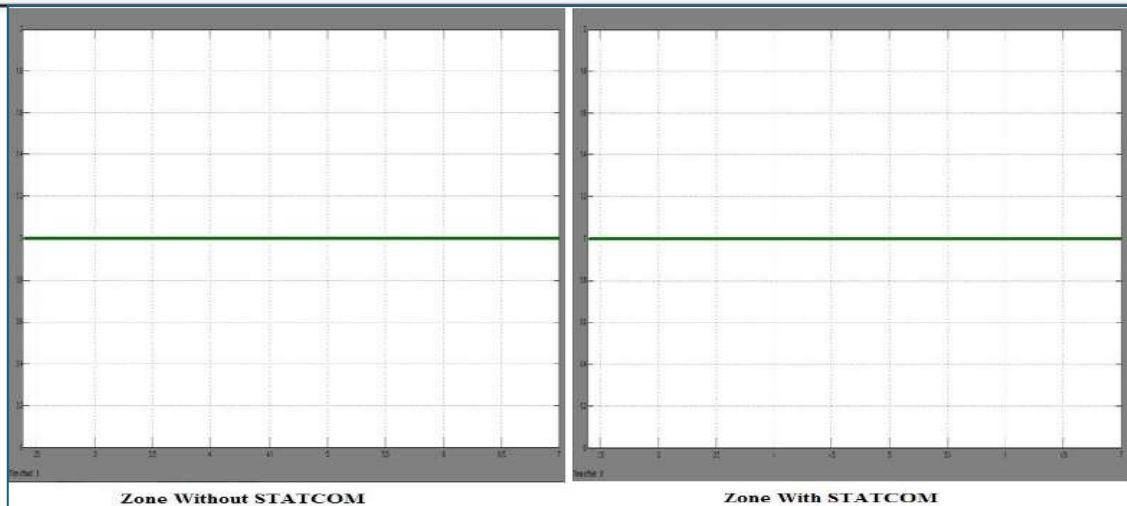


Figure 5.6. With and Without STATCOM Protection Zone Comparison at 78% Line Length.

Reach setting comparison with and without STATCOM AT 78% of T/L length. After simulation, the same impedance relay was tested

and the reach graphs are extracted as in Figure 5.7.

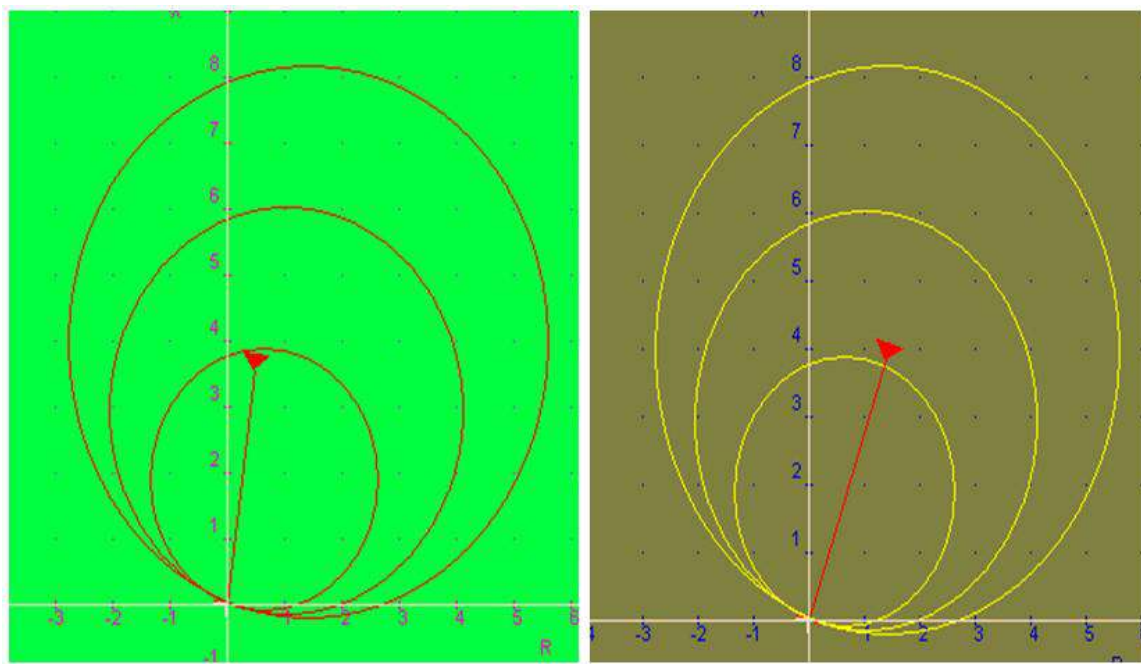


Figure 5.7. Impedance Reach 78% of line with & without STATCOM

- Analysis and remarks**

After faulting at 78% of transmission line length these prominent factors are observed as follow,
 From Simulation, results it is evident that at 78% of T/L distance (60km) i.e. after Midpoint;
 Impedance w/o STATCOM = 3.66Ω
 Impedance with STATCOM = 4.1Ω
 %age Increase in Impedance = 13.66%
 Shifting of Tripping Zone

Variation in the Reach

“UNDER REACHING” observed.

Tripping Time

“Critical Location” of fault.

The impedance is affected by injection of STATCOM in the system, so it can be concluded that STATCOM inside the fault loop will affect the reach of reach. The relay will face Under Reach problem after this critical location.

5.3. Single Phase-Ground Fault Simulation Results

For analysis of the Single phase to Ground fault the simulated model is considered the pre-fault condition are as follows;

Line Voltage= 224KV

Line Current= 600AMPs

Relay Status= Not Operate

After 2 seconds of above pre-fault condition, the 3-Phase fault is simulated in absence of

STATCOM and results are analyzed and then the same fault is injected in presence of STATCOM installed at, 24, 60 and 78% of transmission line length. The system conditions in form of voltage and current are shown in figure 5.6.

The distance relay D60 reach settings are set as;

1st Zone Impedance Reach = 11.79 Ω

2nd Zone Impedance Reach = 18.1 Ω

3rd Zone Impedance Reach = 24.3 Ω



Figure 5.6. System Voltage & Current pre-fault and post-fault conditions.

5.3.1. Single Phase-Ground Fault at 24% of Transmission Line Length

The fault at 24% of transmission line before the mid-point of the line at 24 KM with and without

of STATCOM is simulated. The impedance reach for the relay in both cases is shown in the figure.5.7

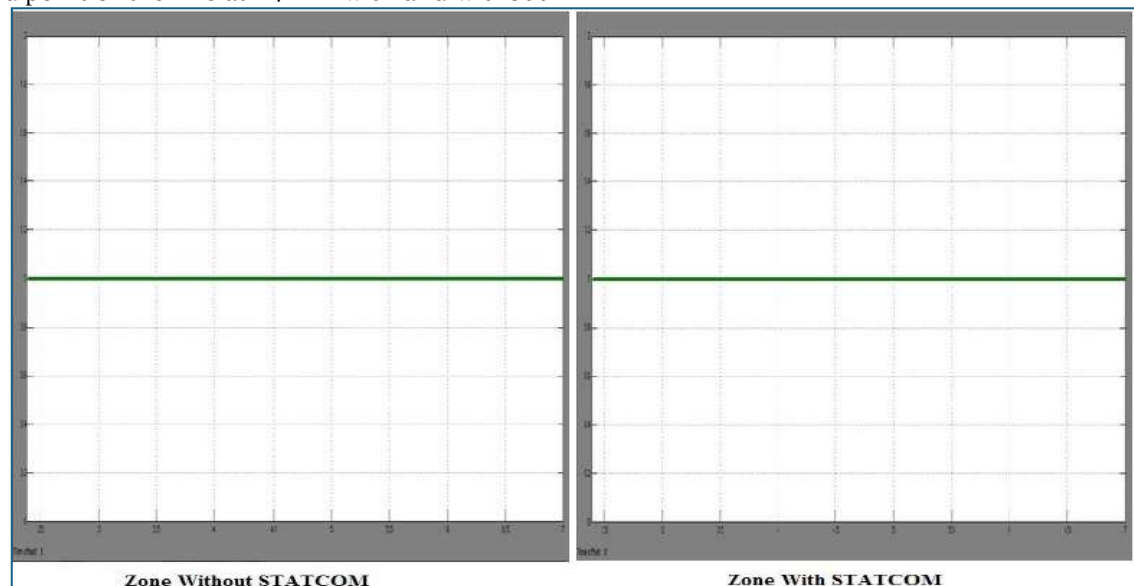


Figure 5.7. With and Without STATCOM Protection Zone Comparison at 24% Line Length.

Reach setting comparison with and without STATCOM AT 24% of T/L length. After

simulation, the same impedance relay was tested and the reach graphs are extracted as in figure 5.8.

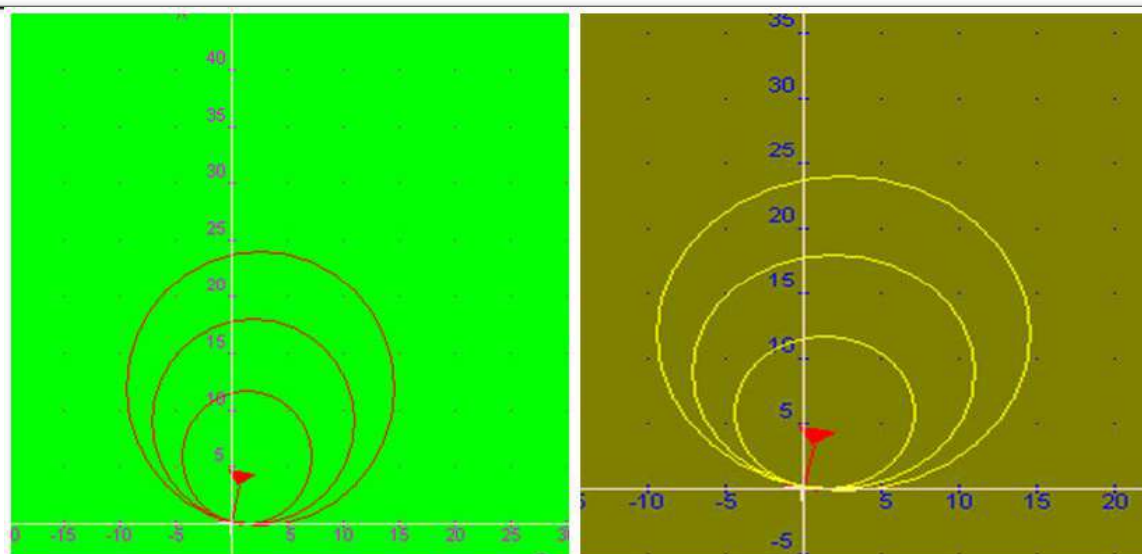


Figure 5.8. Impedance Reach 24% of line with & without STATCOM

- Analysis and remarks**

After Single phase to ground fault at 24% of transmission line length, these prominent factors are observed as follow,

The impedance is not affected by the STATCOM and it is same as shown in figure 5.2.

The relay operates in the same zone and the tripping time is 20 msec.

The same results are extracted from the FREJA testing software.

So it can be concluded that STATCOM outside the fault loop has no impact on the relay performance.

5.3.2. Fault at 60% (60km) of Transmission Line Length

The Single to ground fault at 60% of transmission line before the mid-point of the line at 60 KM with and without of STATCOM is simulated. The impedance reach for the relay in both cases is shown in the Figure.5.9

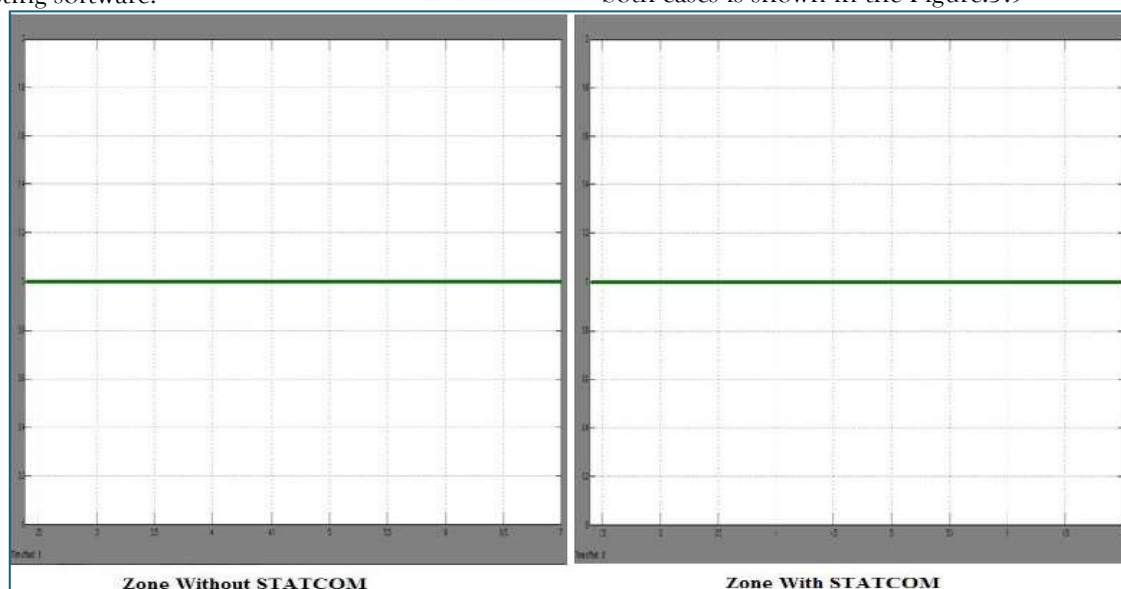


Figure 5.9. With and Without STATCOM Protection Zone Comparison at 60% Line Length.

Reach setting comparison with and without STATCOM AT 60% of T/L length. After simulation, the same impedance relay was tested

and the reach graphs are extracted as in Figure 5.10.

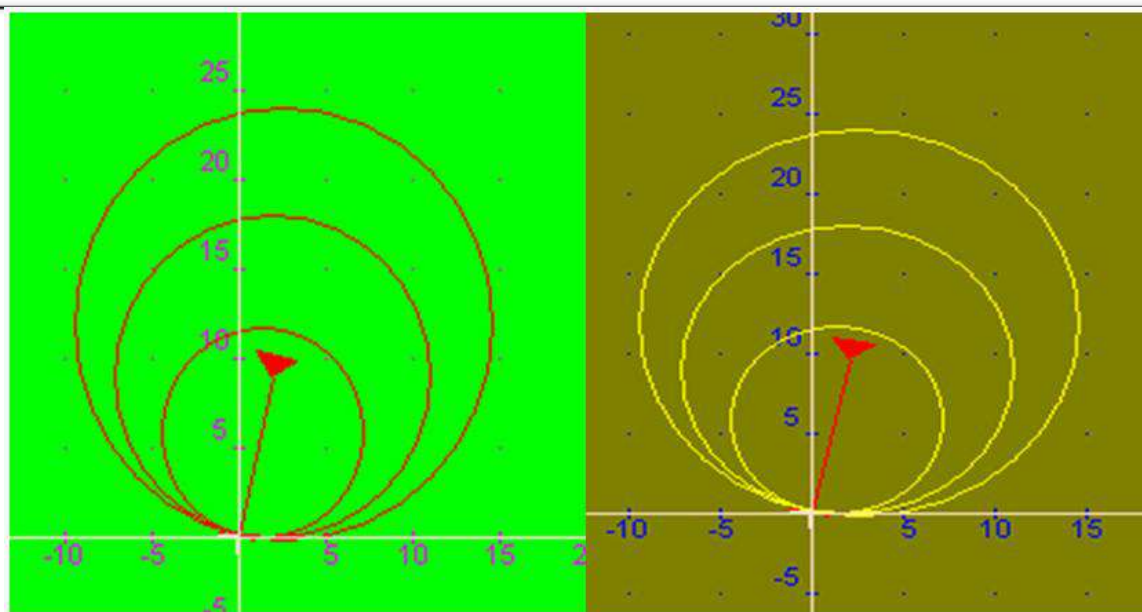


Figure 5.10. Impedance Reach 60% of line with & without STATCOM

- Analysis & remarks**

After Single phase to ground fault at 60% of transmission line length, these prominent factors are observed as follow,

From Simulation results, it is evident that at 60% of T/L distance (60km) i.e after Midpoint;

Impedance without STATCOM is = 9.3 Ω and

Impedance with STATCOM is = 9.85 Ω .

%age Increase in Impedance = 5.9%

Tripping Zone

Reach Settings

It is again supporting to our previous observation. The impedance is affected by injection of STATCOM in the system, so it can be concluded that STATCOM inside the fault loop will affect the reach of reach.

5.3.3. Single Phase to Ground Fault at 78% of Transmission Line Length

The single phase to ground fault at 78% of transmission line before the mid-point of the line at 78 KM with and without of STATCOM is simulated. The impedance reach for the relay in both cases is shown in the figure.5.11

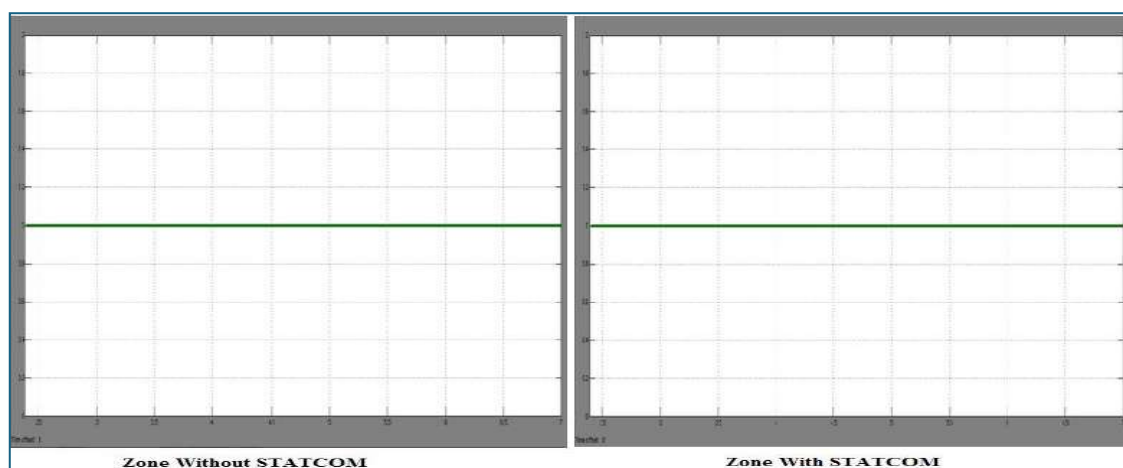


Figure 5.11. With and Without STATCOM Protection Zone Comparison at 78% Line Length.

Reach setting comparison with and without STATCOM AT 78% of T/L length. After simulation, the same impedance relay was tested

and the reach graphs are extracted as in Figure 5.12.

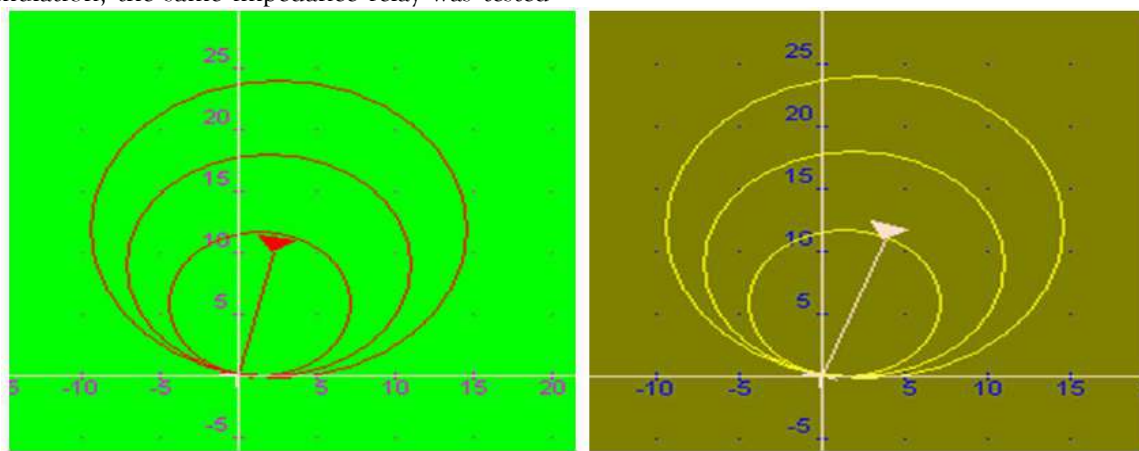


Figure 5.12. Impedance Reach 78% of line with & without STATCOM

• Analysis & remarks

After single phase to ground fault at 78% of transmission line length, these prominent factors are observed as follow,

From Simulation results, it is evident that at 78% of T/L distance (60km) i.e. after Midpoint;

Impedance w/o STATCOM = 10.3 Ω

Impedance with STATCOM = 11.65 Ω.

%age Increase in Impedance = 13.11 %

Tripping zone: changed.

Distance Relay Reach: changed.

“UNDER REACHING” observed.

Tripping Time

“Critical Location” of fault.

The impedance is affected by injection of STATCOM in the system, so it can be concluded that STATCOM inside the fault loop will affect the reach of reach. The relay will face Under Reach problem after this critical location.

5.4. Conclusions and recommendations to eliminate the impact of STATCOM ON RELAY.

- The simulation results depict the following observations.
- The case of the fault in which the STATCOM is out of fault loop the STATCOM has an impact on the impedance protect relay.

- As the STATCOM effects the reactive power of the system so the fault in which the STATCOM is inside the fault loop will change the apparent impedance.

- Due to change in relay reach the relay selectivity and speed of the protection relay will be affected.

- The distance relay will mal-operate and consider the 1st zone fault in 2nd zone, so it will take more time to clear the fault.

As the NTDC is using the standard or basic protection scheme on 220KV lines, so to eliminate the impact of STATCOM on the distance protection scheme the aided communication scheme must be used. To get rid of this under reach problem the Permissive Under Reach Scheme must be used.

5.4.1 Permissive under reach scheme

In presence of STATCOM, the distance relay may under-reach and fail to clear zone-1 faults. To overcome this, a communication-aided “Permissive Under Reach” (PUR) scheme is implemented, where the local relay receives a carrier from the remote relay to operate instantaneously. The relay with PUR is shown in Figure 5.13 for a zone-2 fault.

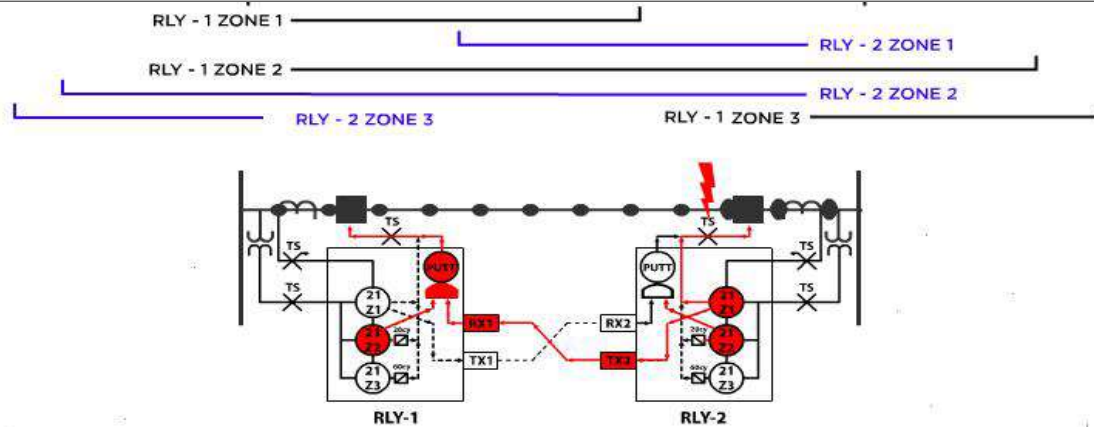


Figure 5.13. Distance Relay Protection with PUR scheme.

The both relays are seeing fault as follow,

- Relay-1 is seeing fault in zone-2
- Relay-2 is seeing fault in zone-1

- The Relay-2 send the carrier signal through a communication channel to Relay-1. The next sequence of operation is shown in figure 5.14.

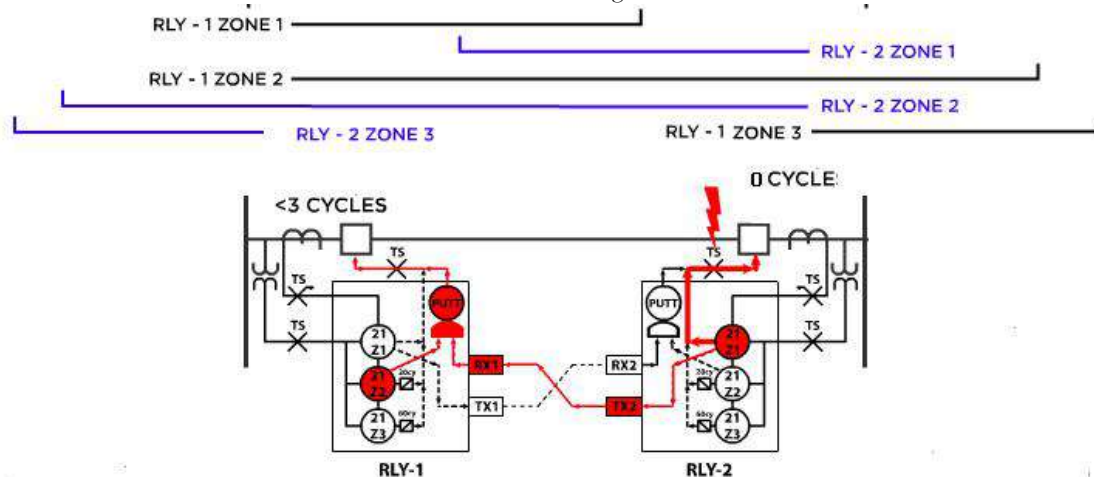


Figure 5.14. Fault Clearance by Distance Relay Protection with PUR scheme.

The Figure 5.14, depicts the following observations,

- Relay-1 sense fault in Zone-2 and received carrier Signal from Remote End relay.
- Relay-1 on is switched to PUR scheme and the Relay-1 trips in zone one time.

- So by PUR, the Relay-1 can compensate it is Under reach Problem. The PUR scheme is simulated in MATLAB as shown in Figure 5.15.

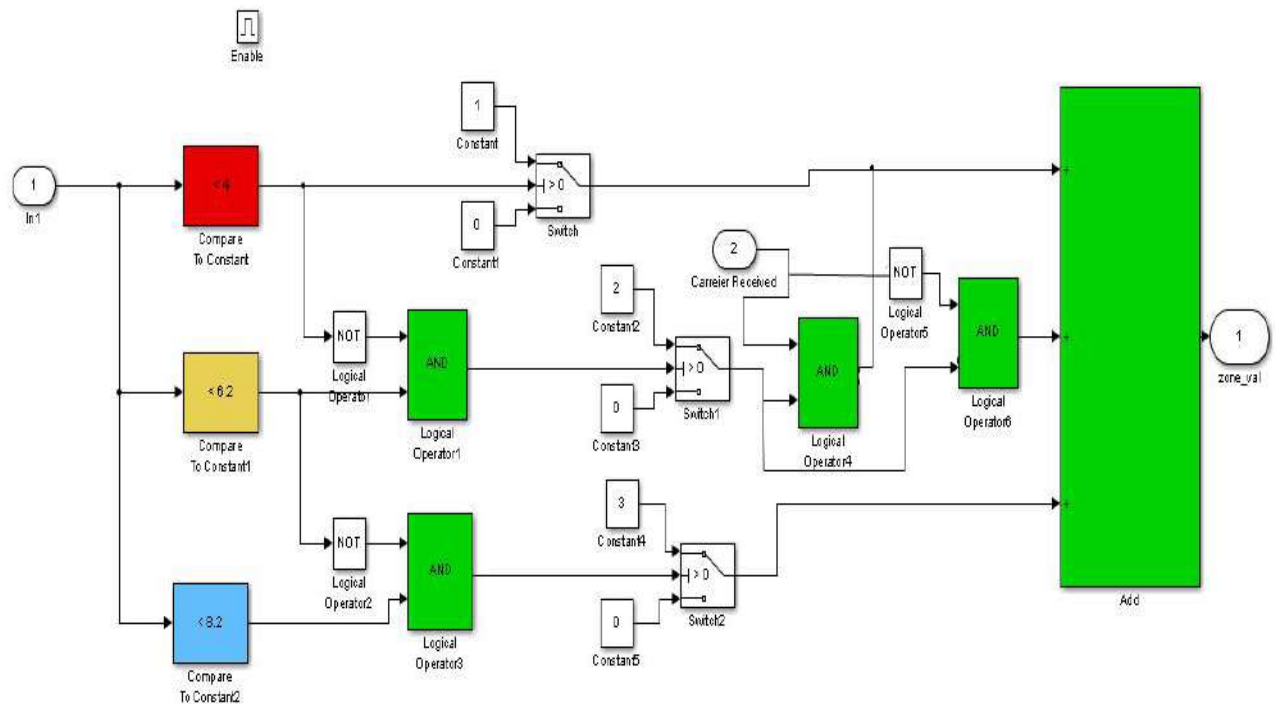


Figure 5.15. Distance Relay Protection PUR scheme model.

From the simulation results;

- Impedance w/o STATCOM = 10.3 Ω
- Impedance with STATCOM = 11.65 Ω .
- %age Increase in Impedance = 13.11 %
- Relay Tripping Time= 20 m sec

So, it is observed that using the PUR Scheme the relay under reach problem can be nullified as a relay is tripping in zone-1 time.

Funding: This research received no external funding.

Data Availability Statement: No data set is used in this study.

Declaration Statement: The authors declare no conflicts of interest.

References

- [1] M. Arun Bhaskar, Dr. S. S Dash, C. Subramani, M. Jagdeesh Kumar, P.R Giresh, M. Varun Kumar. "Voltage Quality Improvement Using DVR", IEEE Transaction on power electronics, 2010, PP. 378- 380..
- [2] W. Hu, Z. Chen, Y. Wang, and Z. Wang, "Flicker Mitigation by Active Power Control of Variable Speed Wind Turbines With Full-Scale Back-to-Back Power Converter," *IEEE Transactions on Energy Conversion*, Vol. 24, No. 3, pp. 640-649, Sept. 2009
- [3] M. P. Bhavaraju, R. Billinton, R. E. Brown, J. Endrenyi, W. Li, A. P. Meliopoulos, and C. Singh, IEEE Tutorial on Electric Delivery System Reliability Evaluation, IEEE Power Engineering Society General Meeting, 2005.
- [4] ERLICH I, RENSCH K, SHEWAREGA F. "Impact of large wind power generation on frequency stability" Proceedings of 2006 IEEE Power Engineering Society General Meeting, Montreal, Canada: 8p.
- [5] L. Tzung-Lin, H. Shang-Hung, and C. Yu-Hung, "Design of D-STATCOM for voltage regulation in microgrids," in *Proc. IEEEECCE*, Sep. 12-16, 2010, pp. 3456-3463.
- [6] Z. Moravey, M. Pazoki and M. Khederazahe, "Impact of UPFC on Power Swing Characteristic and Distance relay Behavior" *IEEE Transactions on Power Delivery*, Vol. 29, No. 1, February 2014.

- [7] Gorakhshanath Abande, M.F.A.R. Satarkar, Mohan Thakre, Dr. V.S Kale, Ganesh Patil, "Impact Analysis of STATCOM on Distance Relay" IEEE 2014, ICIET'14.
- [8]. Mohammed Elsamahy, Sherif Omar Faried and Tarlochan Sidhu, "Impact of Midpoint STATCOM on Generator Loss of Excitation Protection", IEEE Transaction on Power Delivery Vol. 29, No. 2, April 2014.
- [9] www.ntdc.com.pk
- [10]. Mr. Ajaysing, T. Chandan, K. Venkata Rama Mohan, Santhosh Kampelli, Arvind R. Singh, "Advance Distance Protection of Transmission Line in Presence of Shunt Compensator" International Research Journal of Engineering and Technology (IRJET) June 2015.
- [11]. Z. Moravey, M. Pazoki and M. Khederazahe, "Impact of UPFC on Power Swing Characteristic and Distance relay Behavior" IEEE Transactions on Power Delivery, Vol. 29, No. 1, February 2014.
- [12]. Naresh Patnana, Swamynadha Sree Venkata Ramana Chakkirala, "Effect of FACTS Controllers on Impedance Relay Characteristics" International Conference on Engineering Trends and Science & Humanities (ICETSH) 2015.
- [13]. Mohamed Elsamaly, Sherif Omar Faried and Tarlochan Sidhu, "Impact of Midpoint STATCOM on Generator Loss of Excitation Protection" IEEE TRANSACTIONS ON POWER DELIVERY, VOL 29, No. 2, APRIL 2014.
- [14]. Danna Hemasunder, Mohan Thakre, V.S Kale, "Impact of STATCOM on Distance Relay-Modeling and Simulation using PSCAD/EMTDC" IEEE Students Conference on Electrical, Electronics and Computer Sciences 2014.
- [15]. Gorakhshanath Abande, M.F.A.R. Satarkar, Mohan Thakre, Dr. V.S Kale, Ganesh Patil, "Impact Analysis of STATCOM on Distance Relay" International Journal of Innovative Research in Science, Engineering and Technology, Volume 3, Special Issue 3, March 2014.
- [16]. Mohammed Zellague, "A Comparative Study of Impact Series FACTS Devices on Distance Relay in 400KV Transmission Line" Journal of Electrical and Electronics Engineering (JEEE).
- [17]. Mohammed Zellague, Abdelaziz Chaghi, "Impact of TCSC on Distance Protection Setting Based Modified Particle Swarm Optimization Technique" IJISA Vol. 5, May 2013.
- [18]. Gaber El-Saady, Rashad M. Kamel, Essam M. Ali, "Error Analysis in Distance Relay Readings with the presence of FACTS Devices" Innovative System Design and Engineering, Vol. 4, No. 14, 2013.
- [19]. Sankara Subramanian, Anthony Perks, Sarath B Tennakoon, Noel Shammas, "Protection Issues Associated With The Proliferation Of Static Synchronous Compensator (Statcom) Type Facts Devices In Power Systems".
- [20]. Wen-Hao Zhang, Seung-Jae Lee, Myeon-Song Choi and Shigeto Oda, "Considerations on Distance Relay Setting for Transmission Line with STATCOM" IEEE 2010.
- [21]. Mohammad Javad Farah, Behrooz Vahidi, Hossein Askarian Abyaneh, "Effect of FACTS Devices on Measured Impedance by Distance Relay" SINTE 8, 2013.
- [22]. Narain G. Hingorani, Laszlo Gyugyi, "Understanding FACTS" IEEE Power Engineering Society.
- [23]. Yongan Deng, "Reactive Power Compensation of Transmission Lines" Concordia University.
- [24]. Xunchi Wu, "Reactive Power Compensation Based on FACTS Devices" Columbia University.

- [25]. Juan Dixon, Luis Moran, Jose Rodriguez, Ricardo Domke, "Reactive Power Compensation Technologies, State-of-the-Art Review", 2010 (Invited IEEE Paper).
- [26]. Bindeshwar Singh, K.S Verma, Deependra Singh, C.N Singh, Archana Singh, Ekta Agarwal, Rahul Dixit, Baljiv Tyagi, "Introduction to FACTS Controllers A Critical Review" IJRIC Vol. 8, 2011.
- [27]. R. Mohan Mathur, Rajiv K. Varma, "Thyristor-Based FACTS Controllers for Electrical Transmission Systems" IEEE Press.
- [28]. Tariq Masood, R.K. Aggarwal, S.A. Qureshi, R.A.J Khan, "STATCOM Model against SVC Control Model Performance Analysis Techniques".

